EART 22201 Igneous minerals and processes

Mafic and silicic intrusions



Lesson 9

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Mafic and silicic intrusions

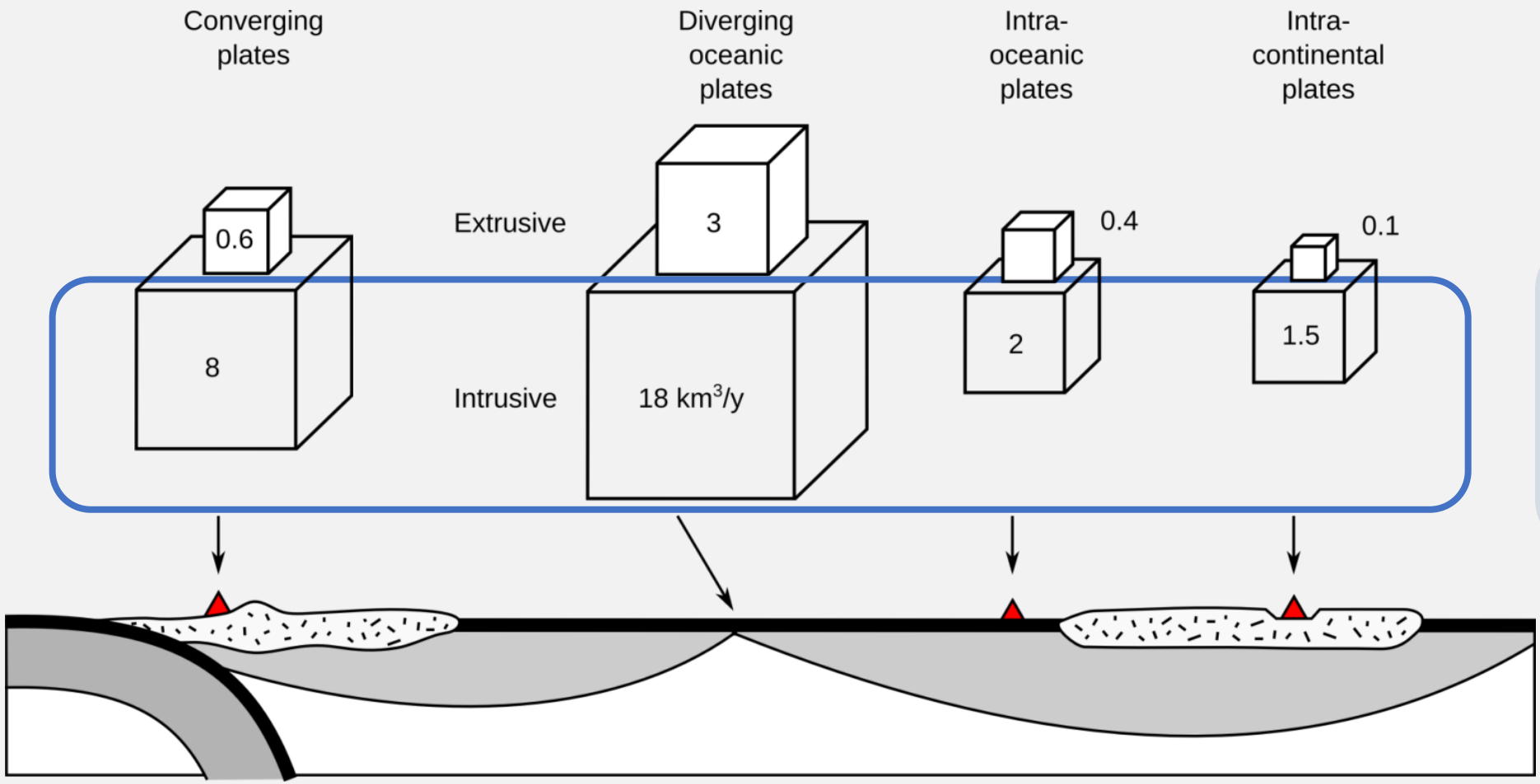
1.1 Igneous intrusions in the geological record

1.2 Why study igneous intrusions?



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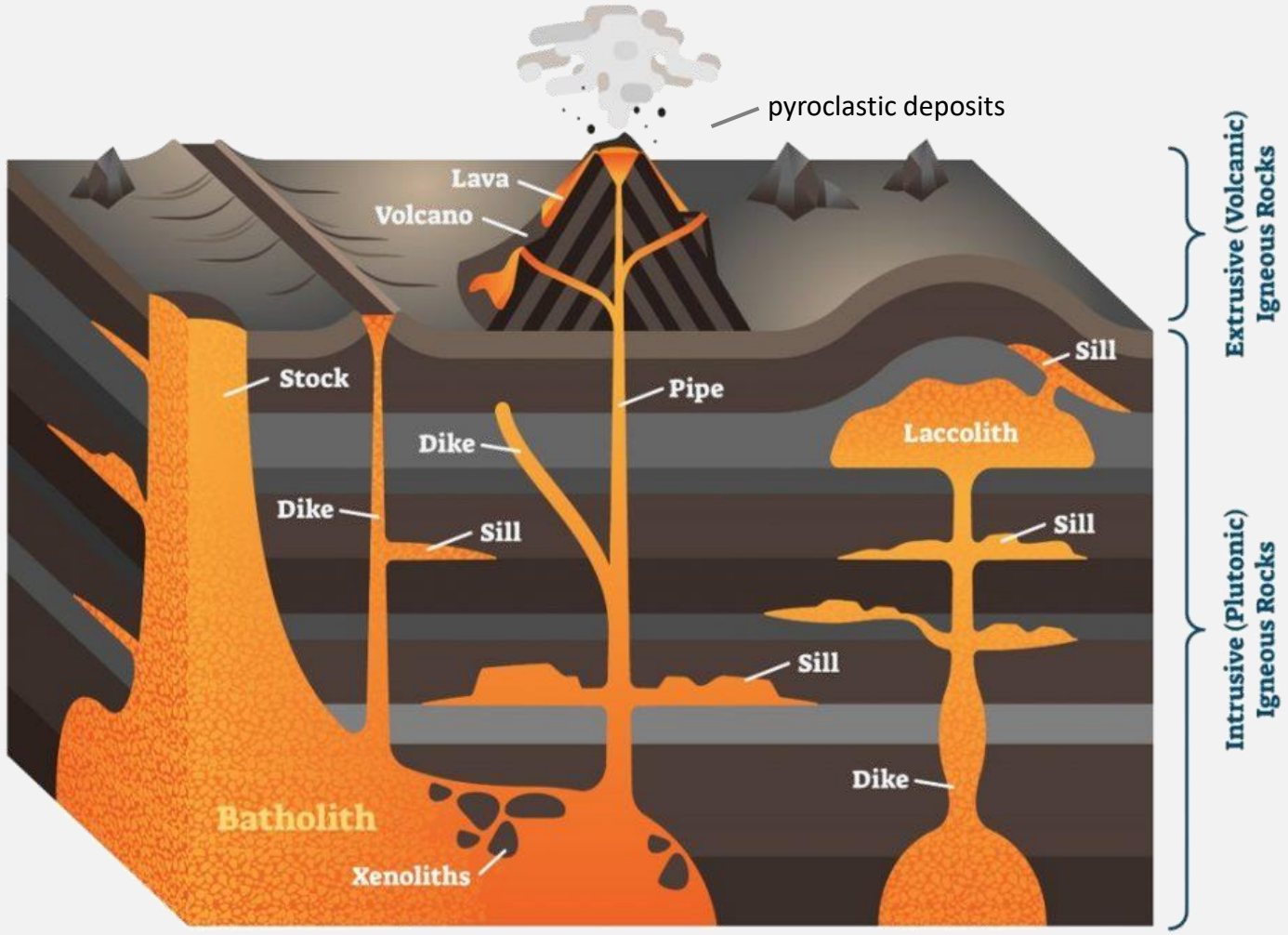
A global inventory of magma production



- ~35% of mean annual magma production is extruded
- **~65% of mean annual magma production is intruded**

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Igneous intrusions of different volumes, scales and geometries



Irregular-shaped intrusions

- Batholith: large – surface exposure >100 km^2
- Stock: small – surface exposure <100 km^2

Regular-shaped intrusions

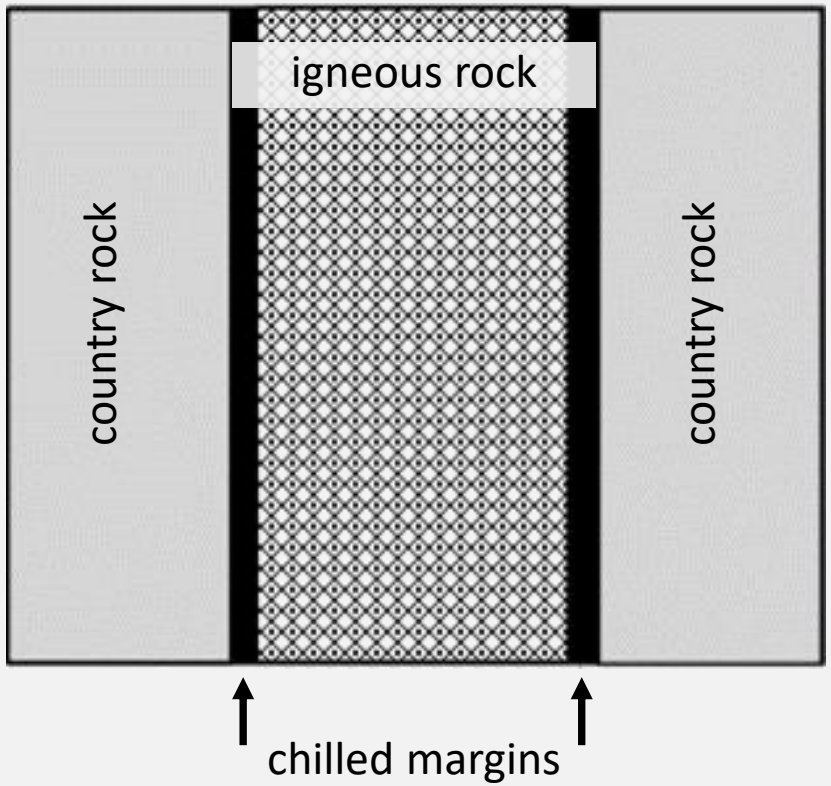
- Laccolith: flat side down, bowl shape
- Lopolith: flat side up, dome shape

Sheet-like intrusions

- Dyke: usually sub-vertical, cross-cutting pre-existing layers of rock. Can extend 10s-100s km
- Sill: usually sub-horizontal, conformable with pre-existing layers of rock

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Intrusions cool inwards from their contact with the country rock

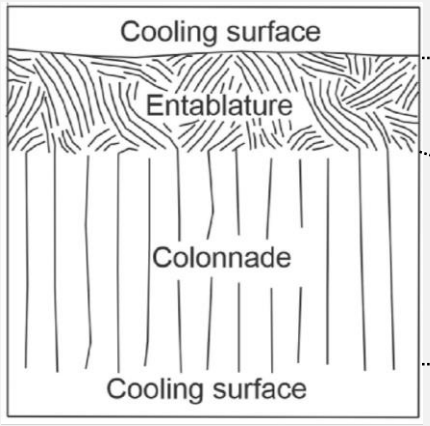




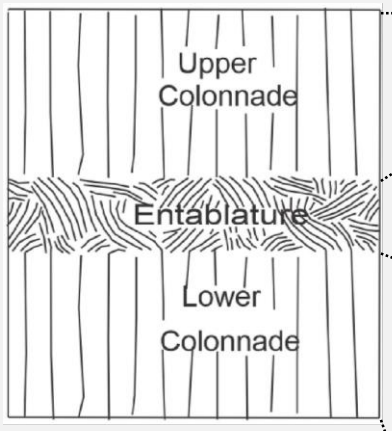
- Chilled margin at the contact with cold country rock
- Grain size increases with distance from the contact
- True for intrusions of all shapes and sizes

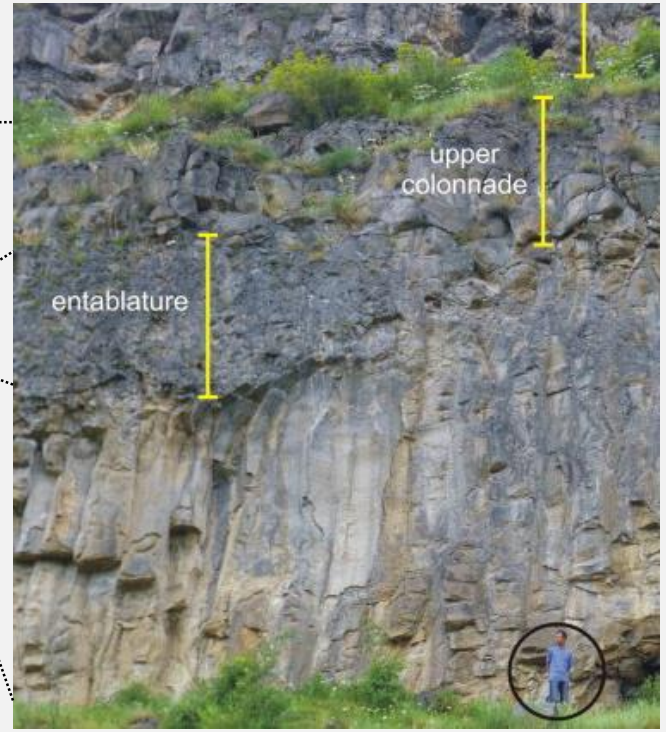
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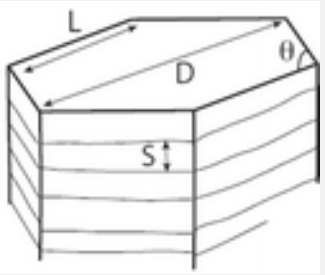
Columnar joints form perpendicular to the cooling surfaces







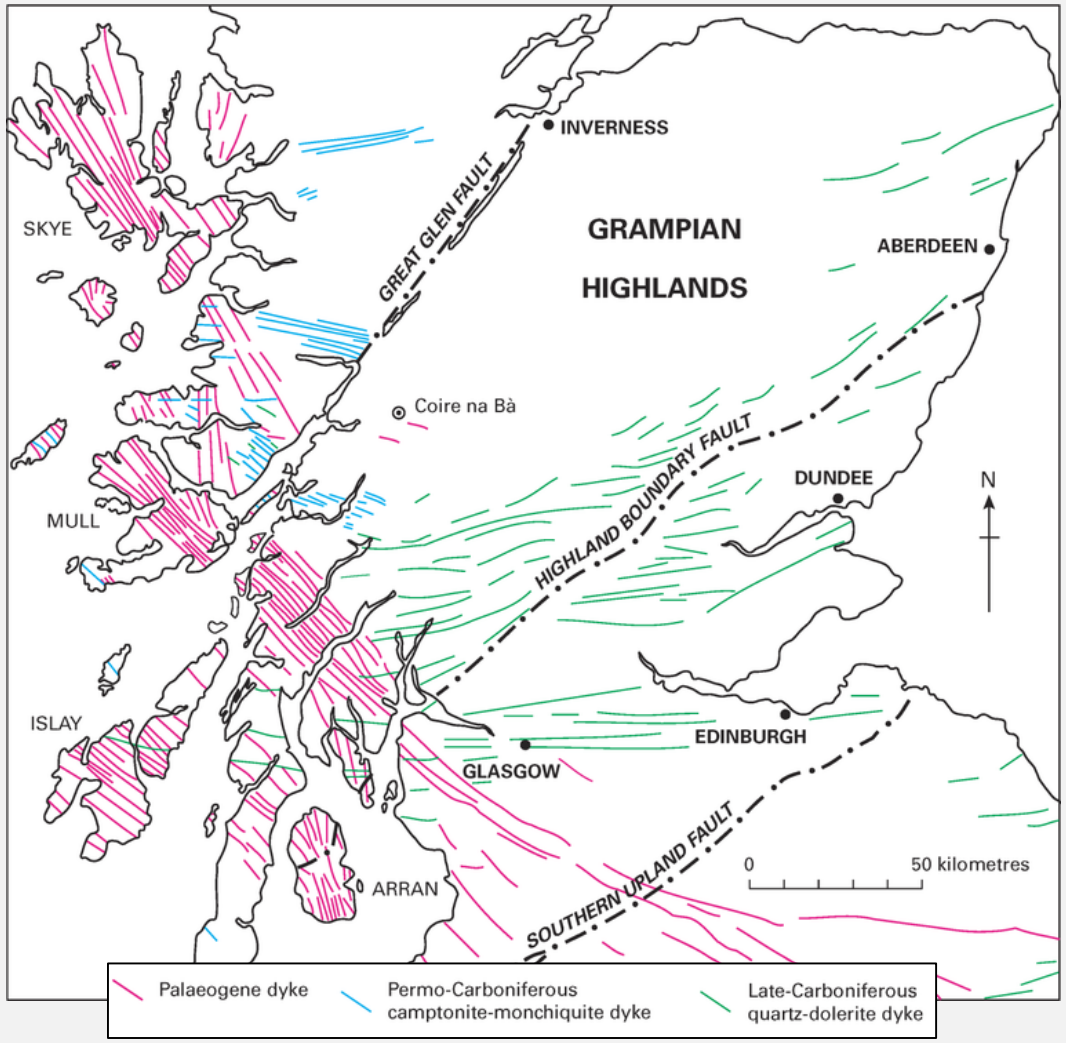




- Joints form after solidification. The rock contracts as it cools, causing cracks (joints) to form
- Columns are commonly hexagonal; number of sides can vary from 3 to 8
- Faster cooling leads to narrower columns

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Cross-cutting field relationships record the relative ages of igneous intrusions



Regional scale:

1. Permo-carboniferous dyke swarm (blue)
2. Late Carboniferous dyke swarm (green)
3. Palaeogene dyke swarm (pink)



Outcrop scale:

1. Light grey gabbro
2. White pegmatite granite
3. Black dolerite

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Mafic and silicic intrusions

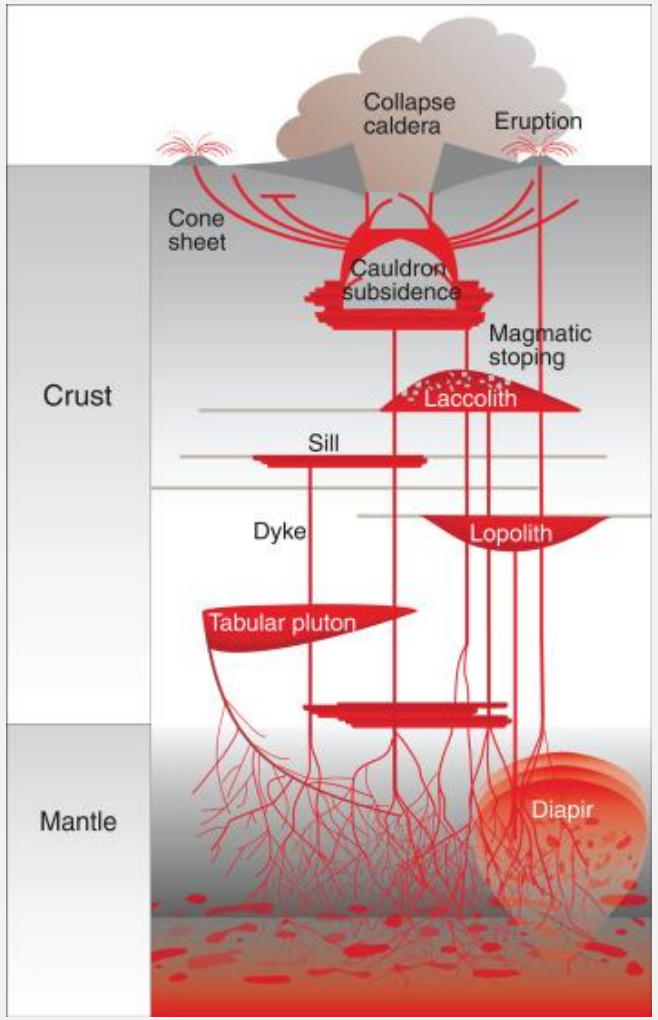
1.1 Igneous intrusions in the geological record

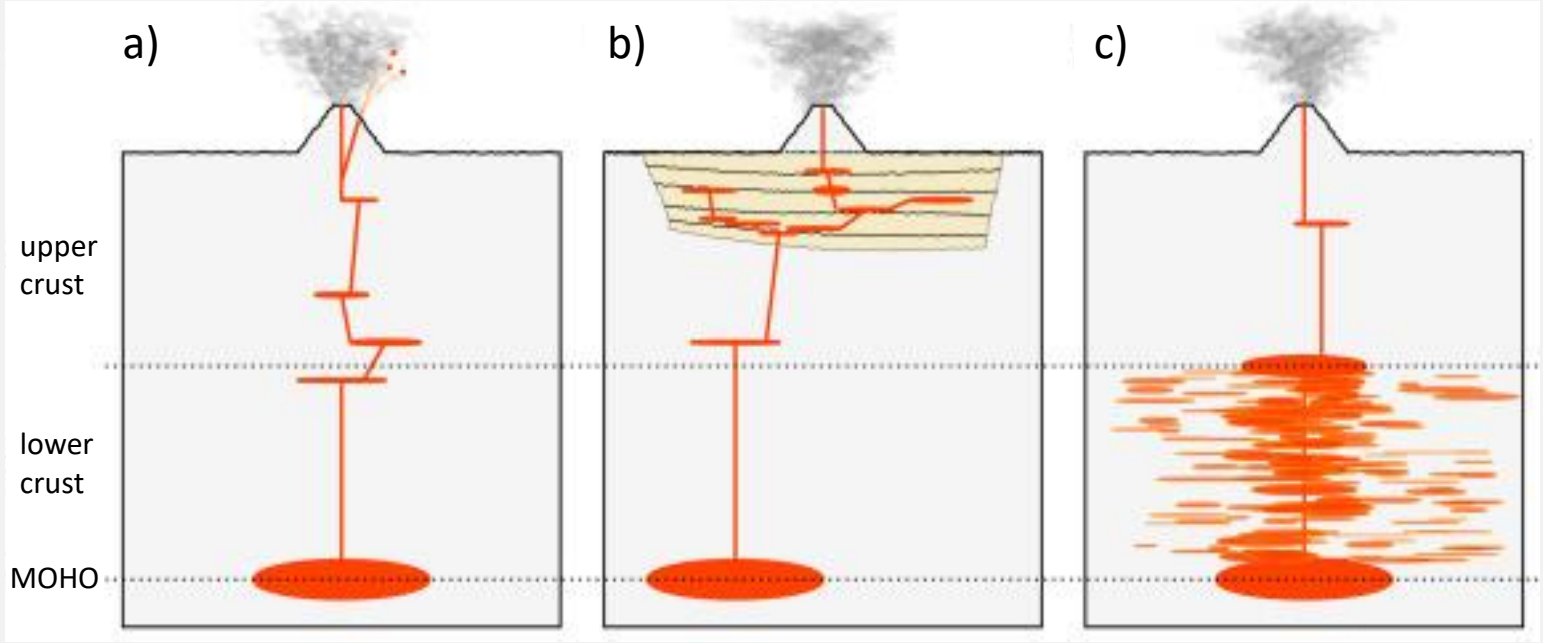
1.2 Why study igneous intrusions?



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Exposed intrusions provide “fossil records” of magma storage and transport in the crust





magmas stored in a deep chamber; limited crustal storage

magmas mainly stored in upper crustal sills

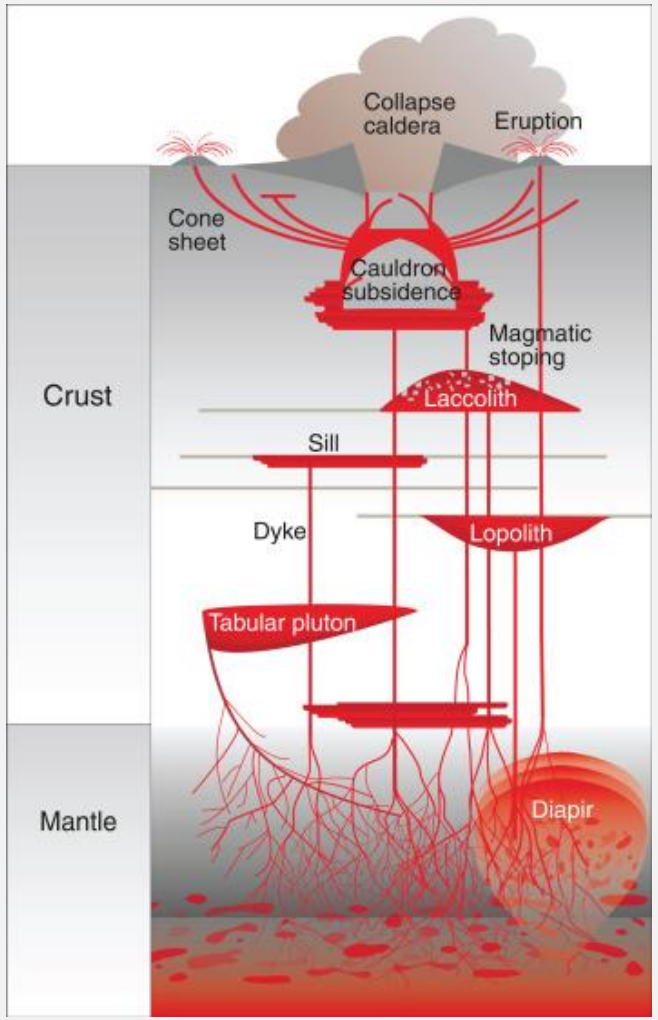
magmas stored in distributed lower crustal sills

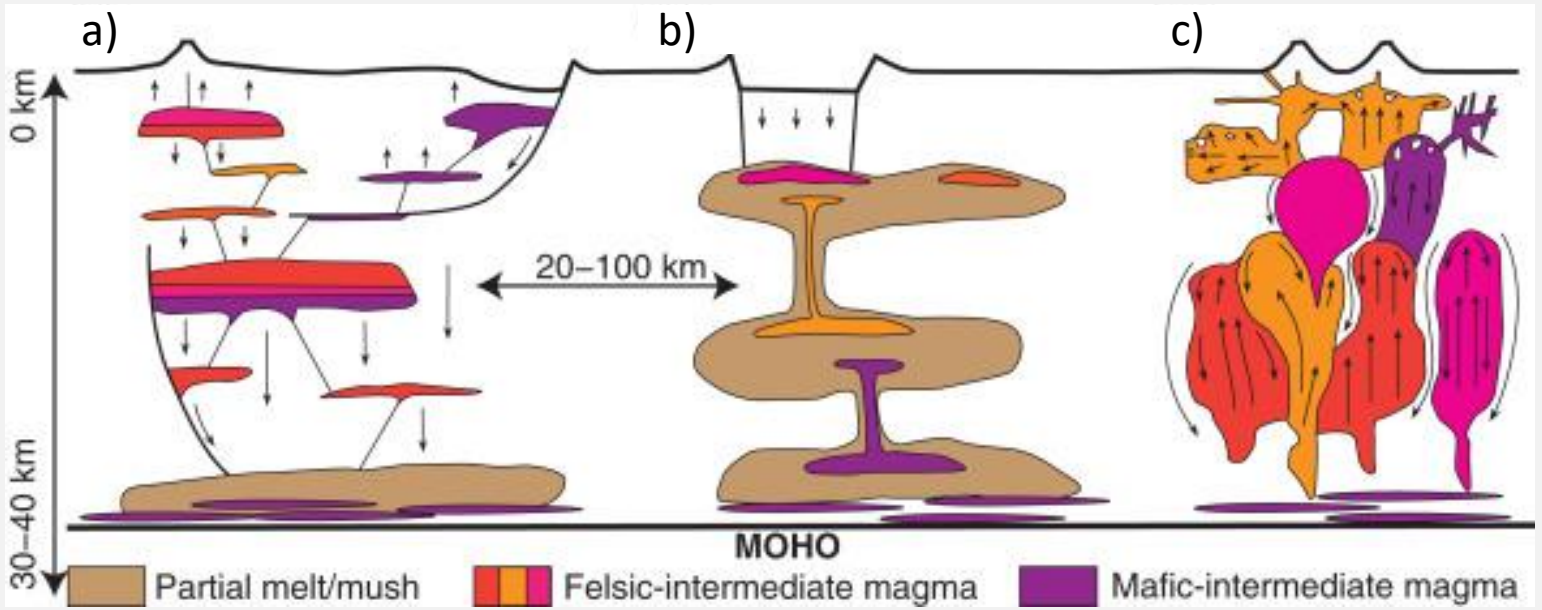
Present-day volcanic hazard modelling relies on knowledge of:

- Where are magmas stored: deep or shallow? How does this vary with composition?
- The magma ascent pathway to the surface: slow or fast? With or without storage?

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Exposed intrusions provide “fossil records” of magma storage and transport in the crust





Tabular intrusions; magmas ascend through dykes that form by fracturing the crust

Mush: crust composed of a rigid crystal framework; melts ascend through the pore space

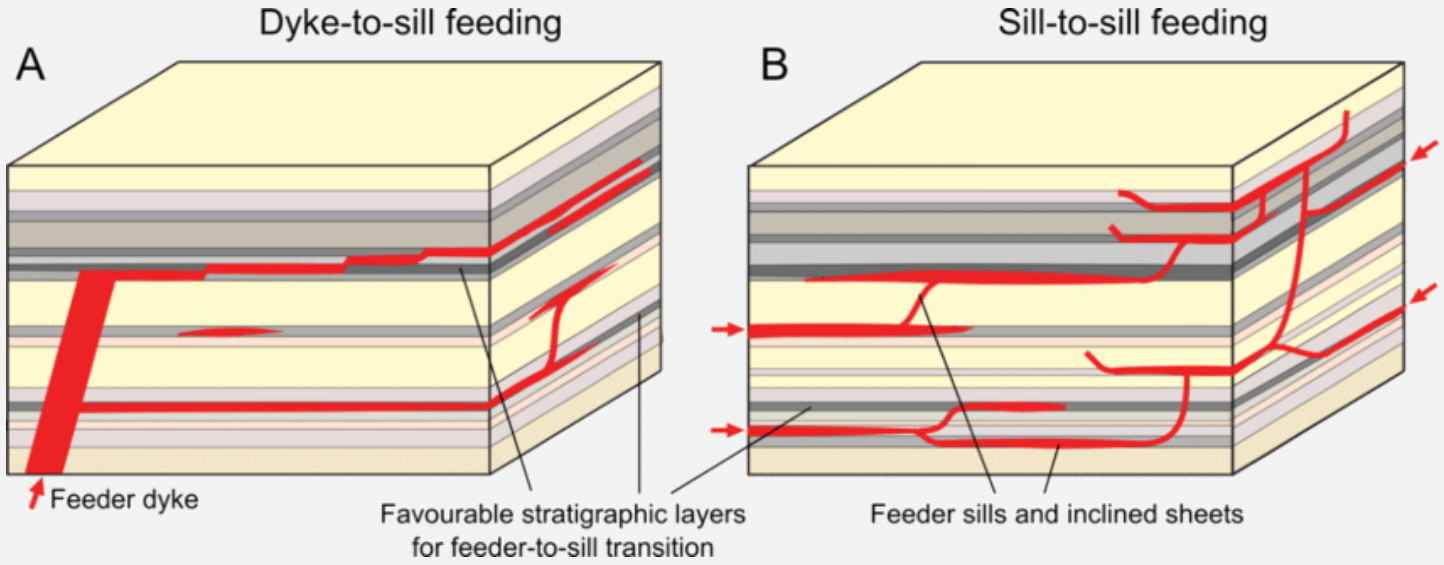
Diapirs: magmas ascend as large masses, driven by buoyancy

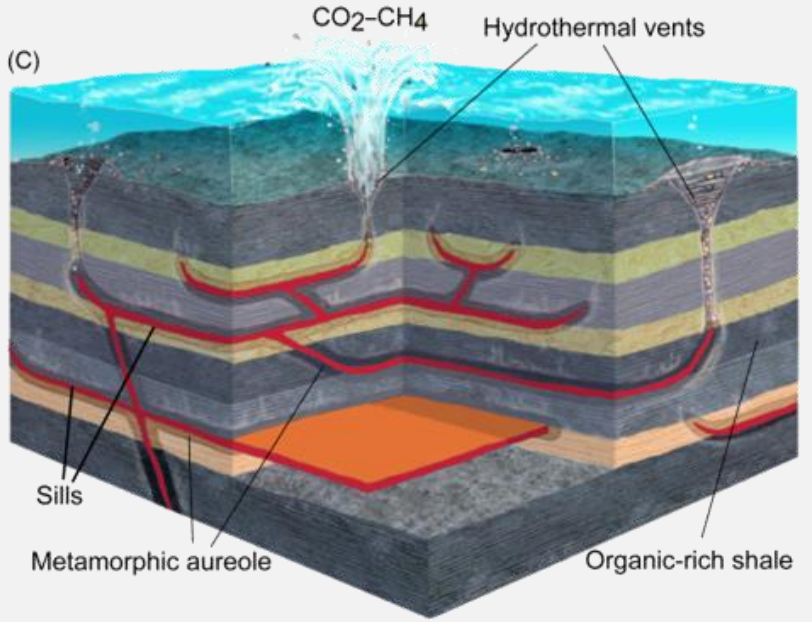
Present-day volcanic hazard modelling relies on knowledge of:

- Where are magmas stored: deep or shallow? How does this vary with composition?
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Subsurface networks of intrusions can control fluid flow pathways in the crust





- Intrusion geometry often constrained by the pre-existing geology
- Intrusion emplacement can enhance permeability by fracturing the intruded crustal rocks
- Solidified intrusions themselves usually have low permeability

- hydrocarbon reservoirs
- carbon capture and storage
- geothermal energy
- formation of ore deposits

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Mafic and silicic intrusions

You should now be able to:

- Recognise the terminology used to describe different forms of igneous intrusion
- Describe the cooling and solidification features exhibited by igneous intrusions
- Discuss how intrusions now exposed at the Earth's surface allow interpretation of how magmas are stored and transported in the crust
- Outline why knowledge of igneous intrusion geometry is useful in industrial and applied geological contexts



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Mafic and silicic intrusions

2.1 Mafic intrusions: gabbro and dolerite

2.2 Closed-system fractional crystallization in a large mafic magma chamber: the Skaergaard layered intrusion

2.3 Crystallization in smaller mafic intrusions: the Palisades Sill



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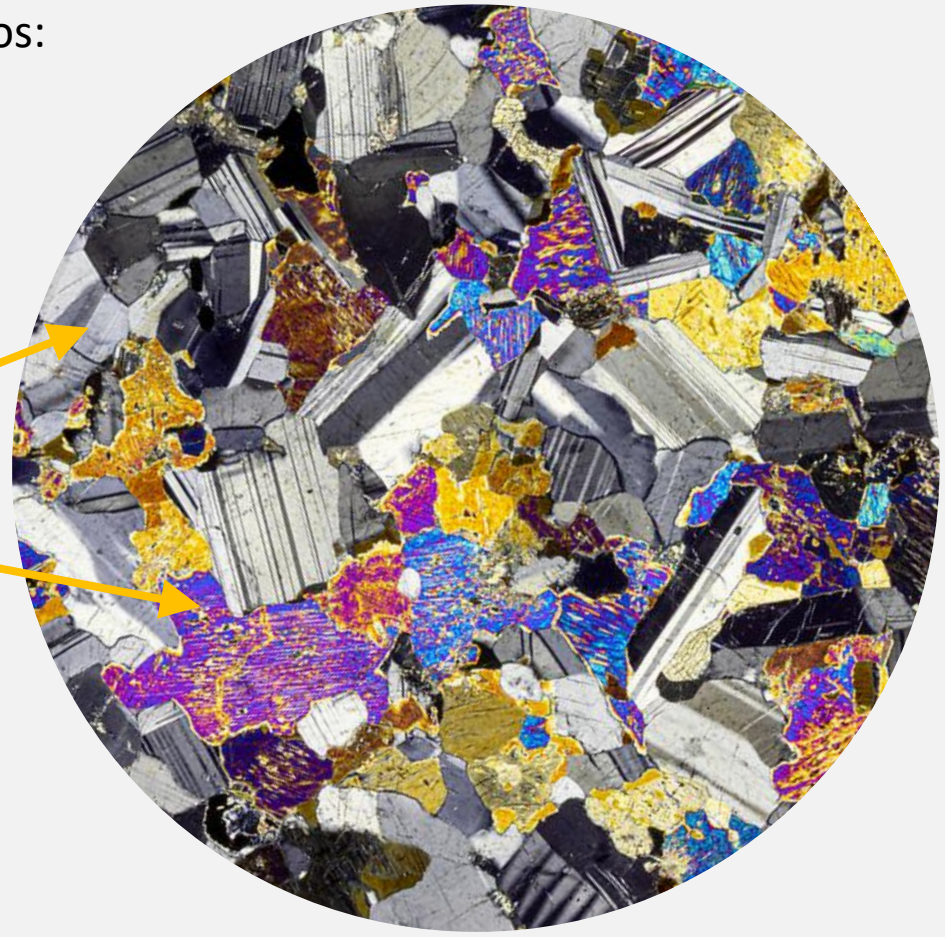
Gabbro: what to look for in a thin section

Medium-grained equivalent is dolerite (sometimes called diabase, especially in American literature)

Essential minerals found in <u>all</u> gabbros:

- Clinopyroxene (augite)
- Plagioclase (high-Ca)

Same essential minerals as basalt



Coarse grain size. Possible textures: cumulate, interstitial, ophitic, poikilitic

Type minerals in tholeiitic gabbros:

- Olivine
- Oxides e.g. magnetite, spinel

Type minerals in alkali gabbros:

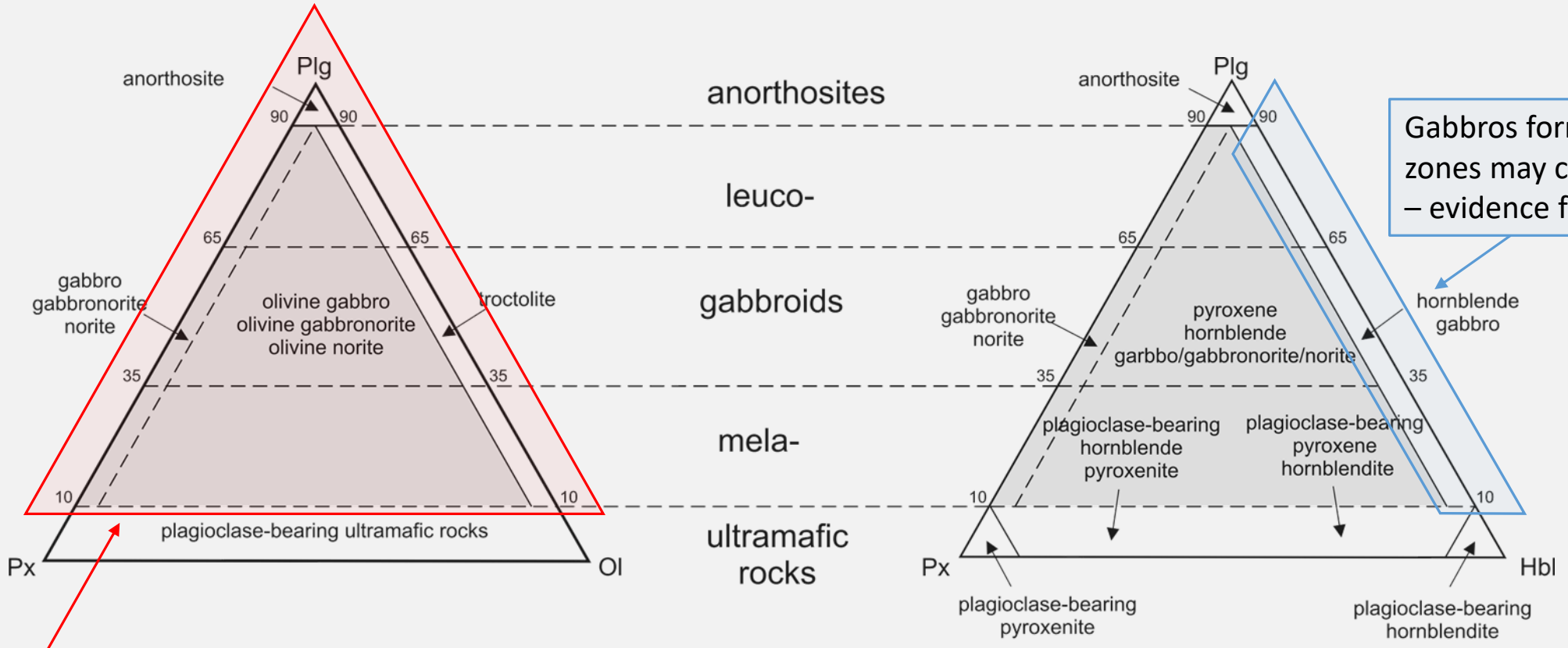
- Olivine
- Nepheline*
- Oxides e.g. magnetite, spinel

Same type minerals as basalt. Type minerals may be found in some gabbros, but not necessarily all.

*We will not look at nepheline-bearing rocks in this unit

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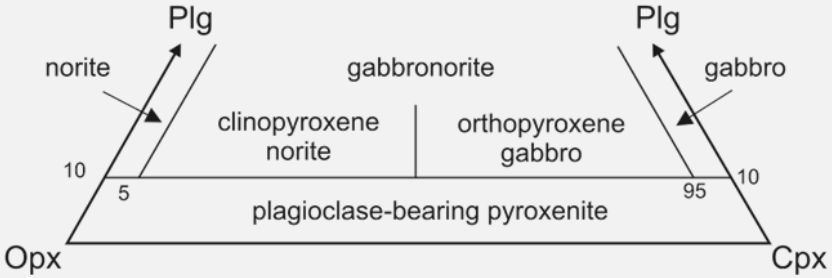
Gabbros can be classified in further detail on the basis of their modal mineralogy



Gabbros formed in subduction zones may contain **hornblende** – evidence for wet magma

Typical gabbro mineralogy is **plag + cpx ± ol**

No hydrous minerals – evidence for dry magma



These diagrams are in the Practical Reference Handout

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Mafic and silicic intrusions

You should now be able to:

- List the essential minerals and type minerals for gabbros and dolerites
- Classify gabbros on the basis of their modal mineralogy using an appropriate ternary diagram
- Interpret the tectonic origin of a gabbro based on the presence or absence of hydrous minerals



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Mafic and silicic intrusions

2.1 Mafic intrusions: gabbro and dolerite

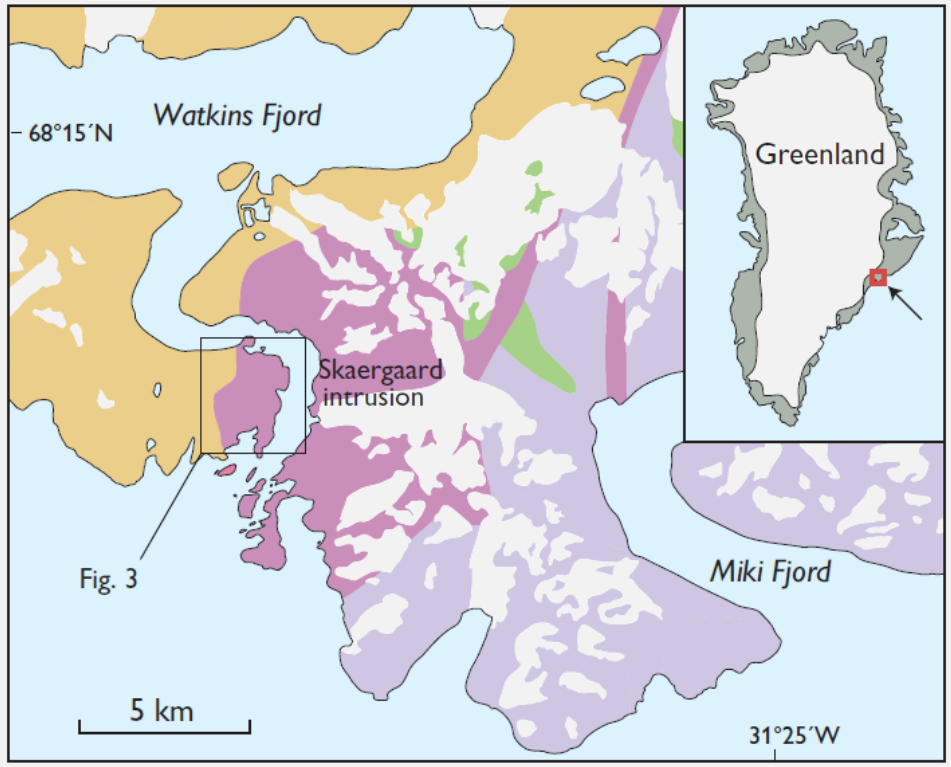
2.2 Closed-system fractional crystallization in a large mafic magma chamber: the Skaergaard layered intrusion

2.3 Crystallization in smaller mafic intrusions: the Palisades Sill



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Crystallization in a fossil magma chamber: The Skaergaard layered intrusion

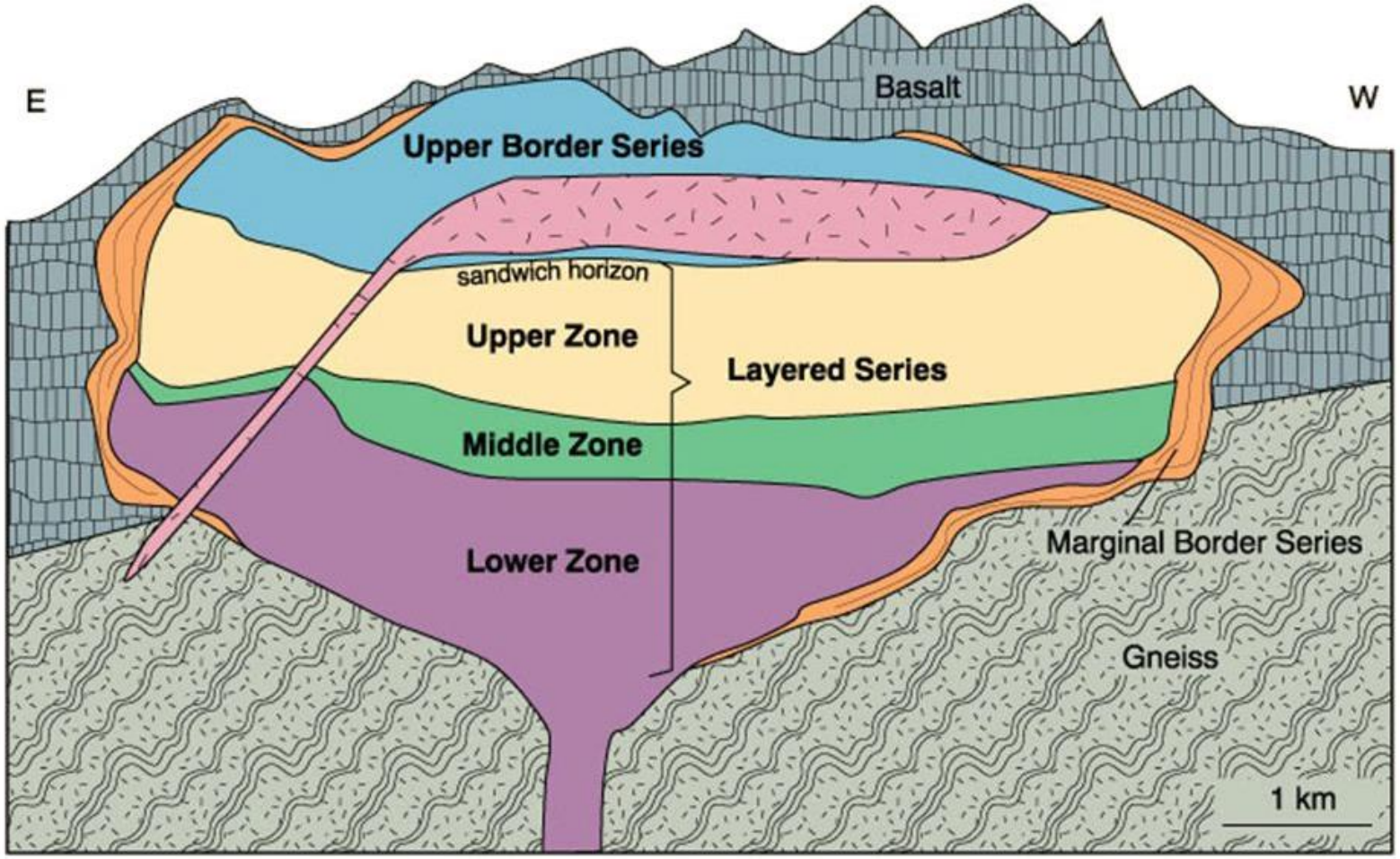


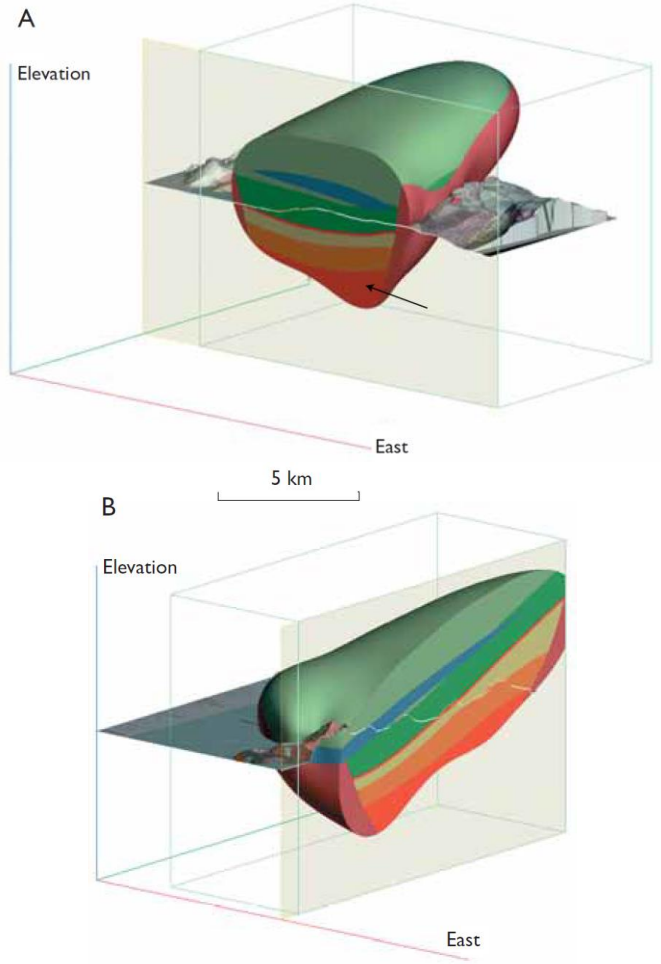


Best (2006) Section 12.4.2
Gill (2010) Chapter 4

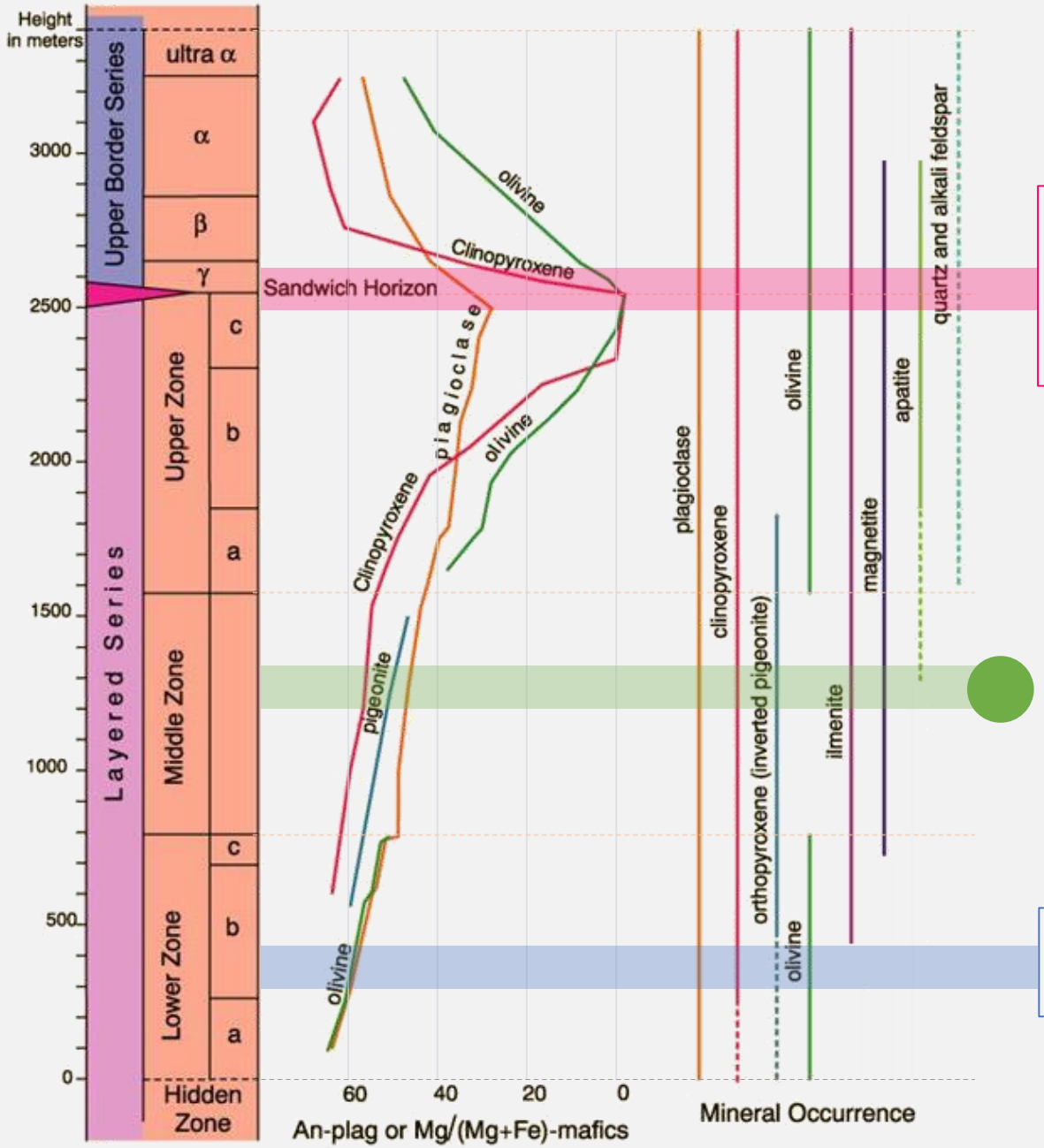
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Stratigraphy of the Skaergaard intrusion





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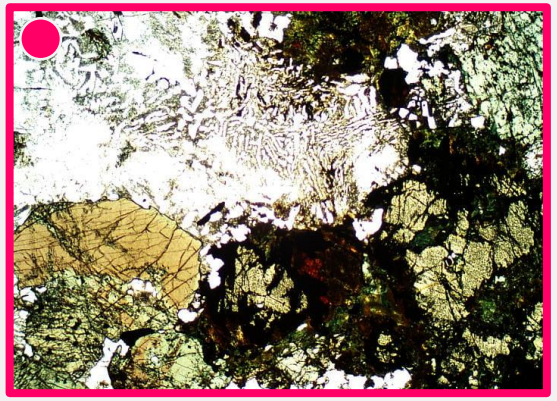
Augite (green and brown), fayalite, plagioclase, oxides, apatite, myrmekite*

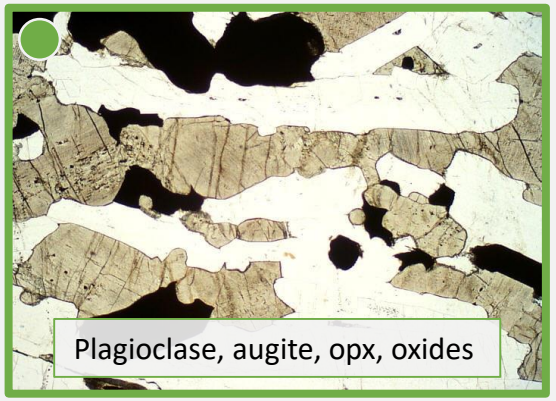
*fine intergrowth of quartz rods inside a plagioclase crystal

Modal layering:

The modal proportions of minerals change with height in the intrusion

Plagioclase, augite, olivine, minor oxides

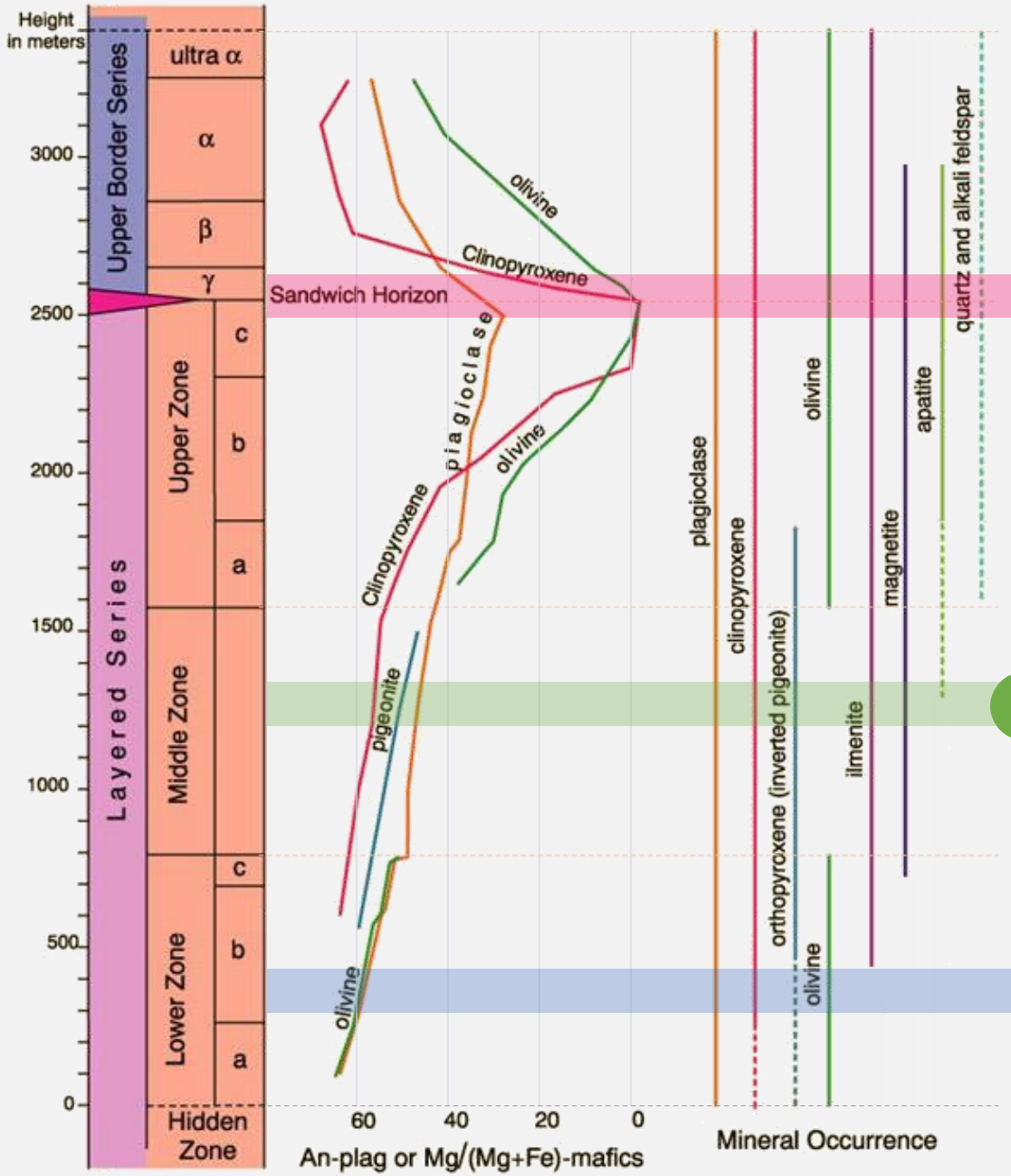




Plagioclase, augite, opx, oxides



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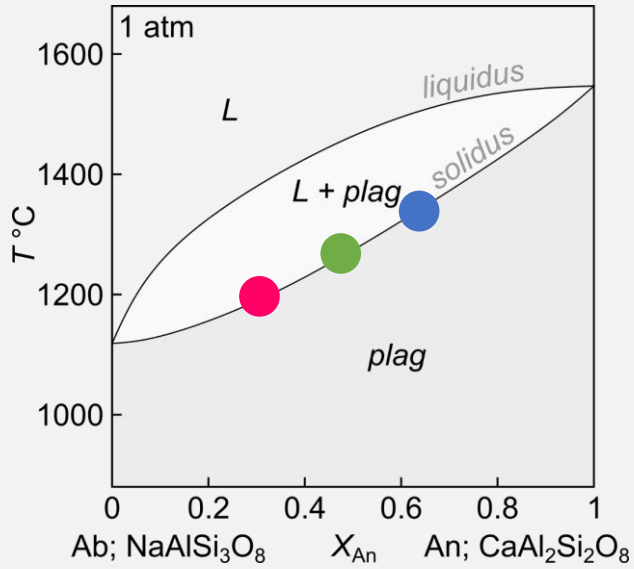


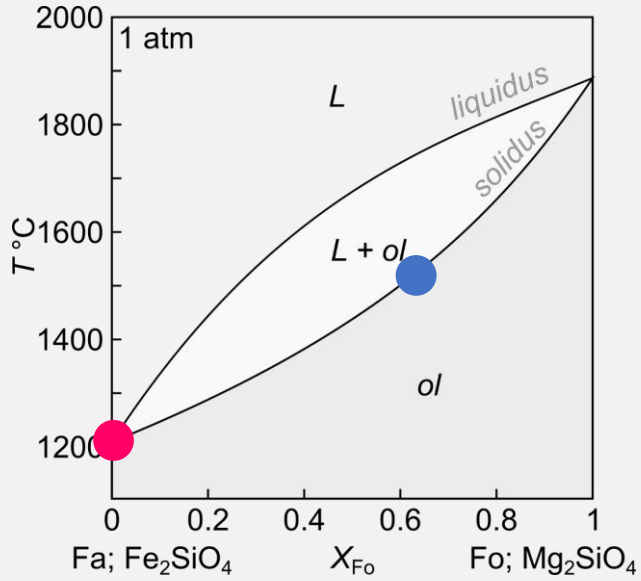
An_{30} plagioclase
Fo_0 olivine
Fe-rich clinopyroxene

Cryptic layering:

The composition of each mineral changes with height in the intrusion

An_{65} plagioclase
Fo_{65} olivine
Mg-rich clinopyroxene





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Skaergaard layered gabbros









Rhythmic layering:

A repeated layering pattern common in mafic layered intrusions

- Bases of layers are dominated by dense ferromagnesian phases (olivine, pyroxene)
- Tops of layers are dominated by less dense silica-rich phases, mainly plagioclase
- “Magmatic sedimentation” e.g. normal grading in layers

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Mafic and silicic intrusions

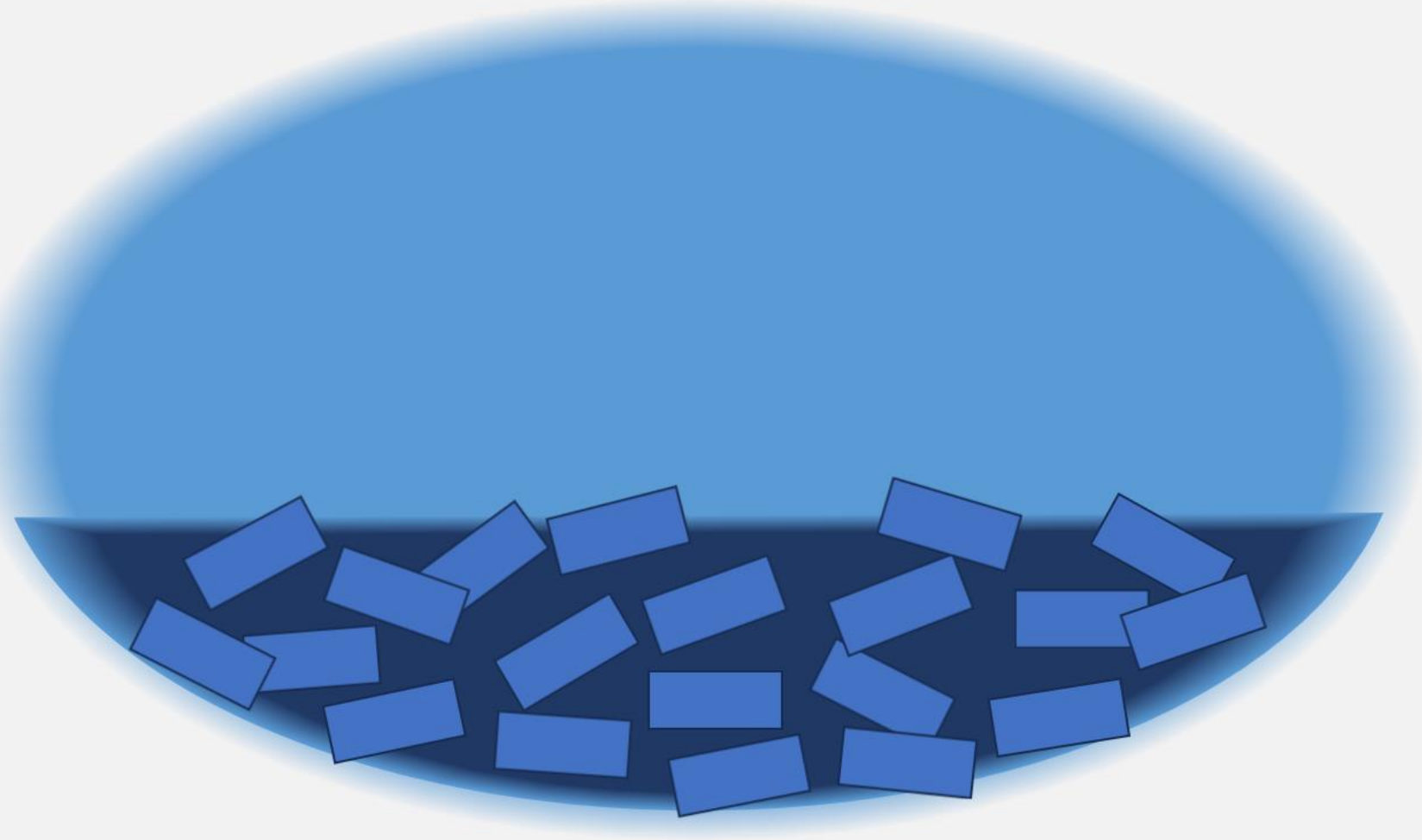
You should now be able to:

- Describe the different types of layering (modal, cryptic, rhythmic) that occur in large mafic intrusions
- Explain why modal and cryptic layering occur in large mafic intrusions, with reference to closed-system fractional crystallization



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“Magmatic sedimentation” in large mafic intrusions creates cumulate textures



1. Large, euhedral crystals grow free-floating in the magma

2. Dense crystals settle through the magma and accumulate at the floor of the chamber

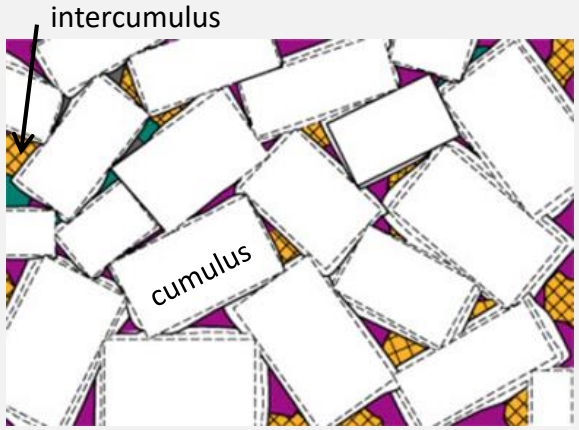
Cumulus minerals

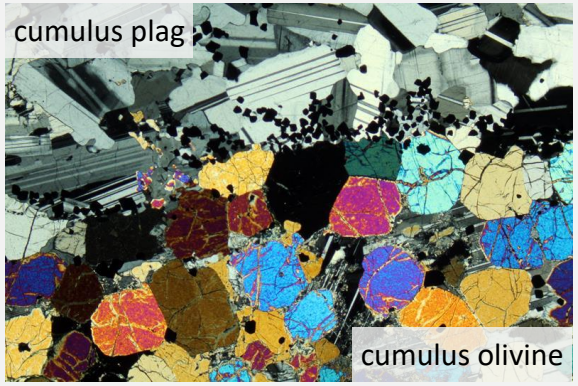
3. Minerals crystallize in the pore spaces between the cumulus phases

Intercumulus minerals

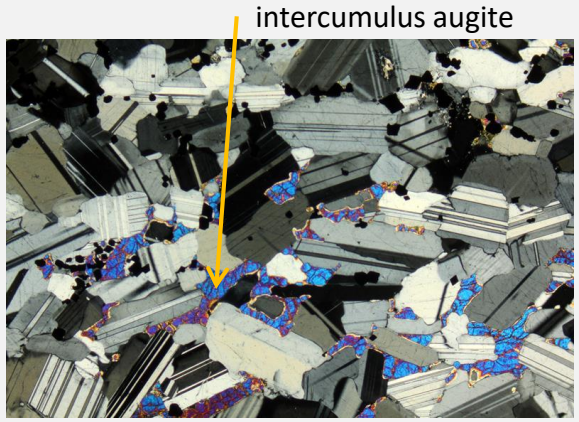
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“Magmatic sedimentation” in large mafic intrusions creates cumulate textures

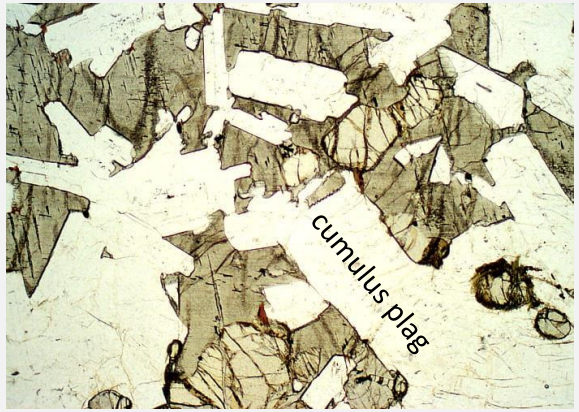




Rum, Scotland



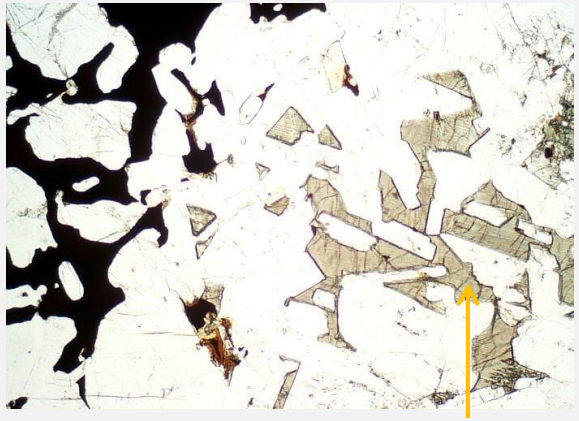
Rum, Scotland



Skaergaard



Skaergaard



Skaergaard intercumulus augite

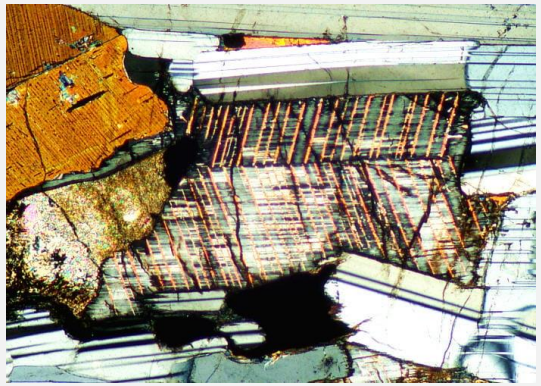
What to look for:

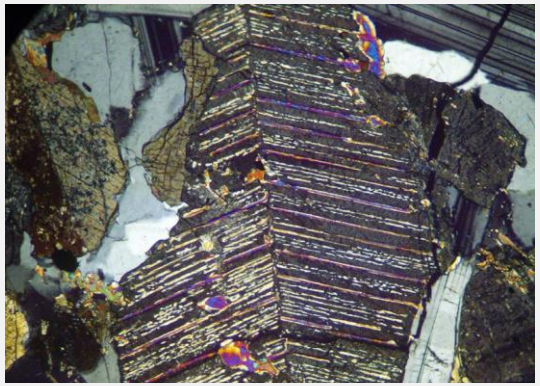
Cumulus phase(s):
- Touching framework of euhedral crystals
- Preferred mineral alignment
- Crystals completely surrounded by another mineral

Intercumulus phase(s):
- Anhedral crystals filling pore spaces – sometimes in optical continuity
- Crystals that fully enclose other euhedral minerals

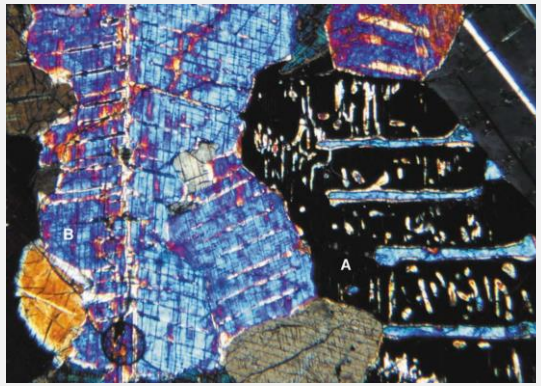
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Slow cooling in large mafic intrusions may cause exsolution in pyroxenes



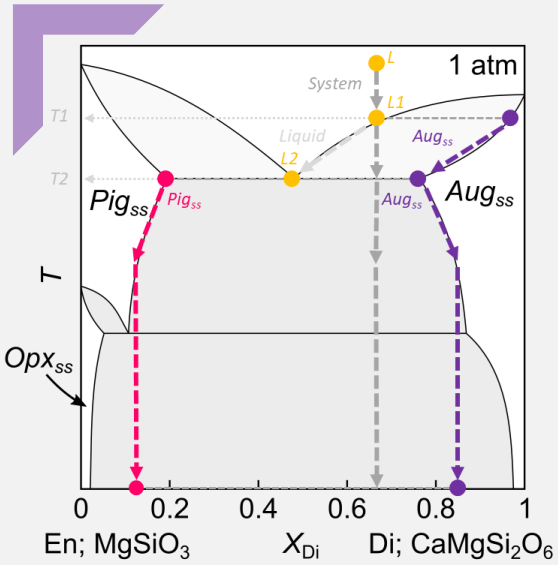


Slow cooling: one generation of thin exsolution lamellae



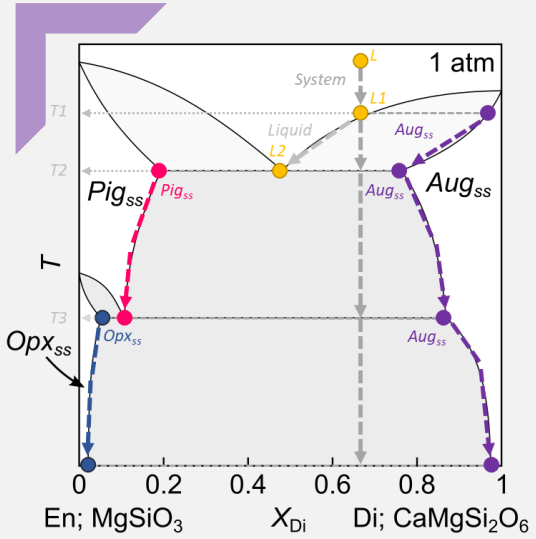


Ultra-slow cooling: thick or blebby exsolution lamellae, and/or "inverted pigeonite" with two generations of exsolution lamellae



Phase diagrams (extra) are from EART 26201 Principles of Geochemistry, Lecture 4

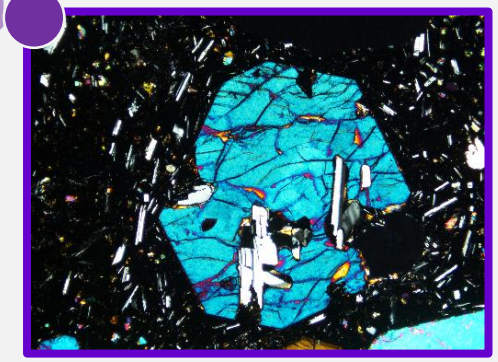
Key point: Exsolution textures are an indicator of slow cooling



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Exsolution in pyroxenes - which pyroxene grows where?

From EART 26201 Principles of Geochemistry, Lecture 4



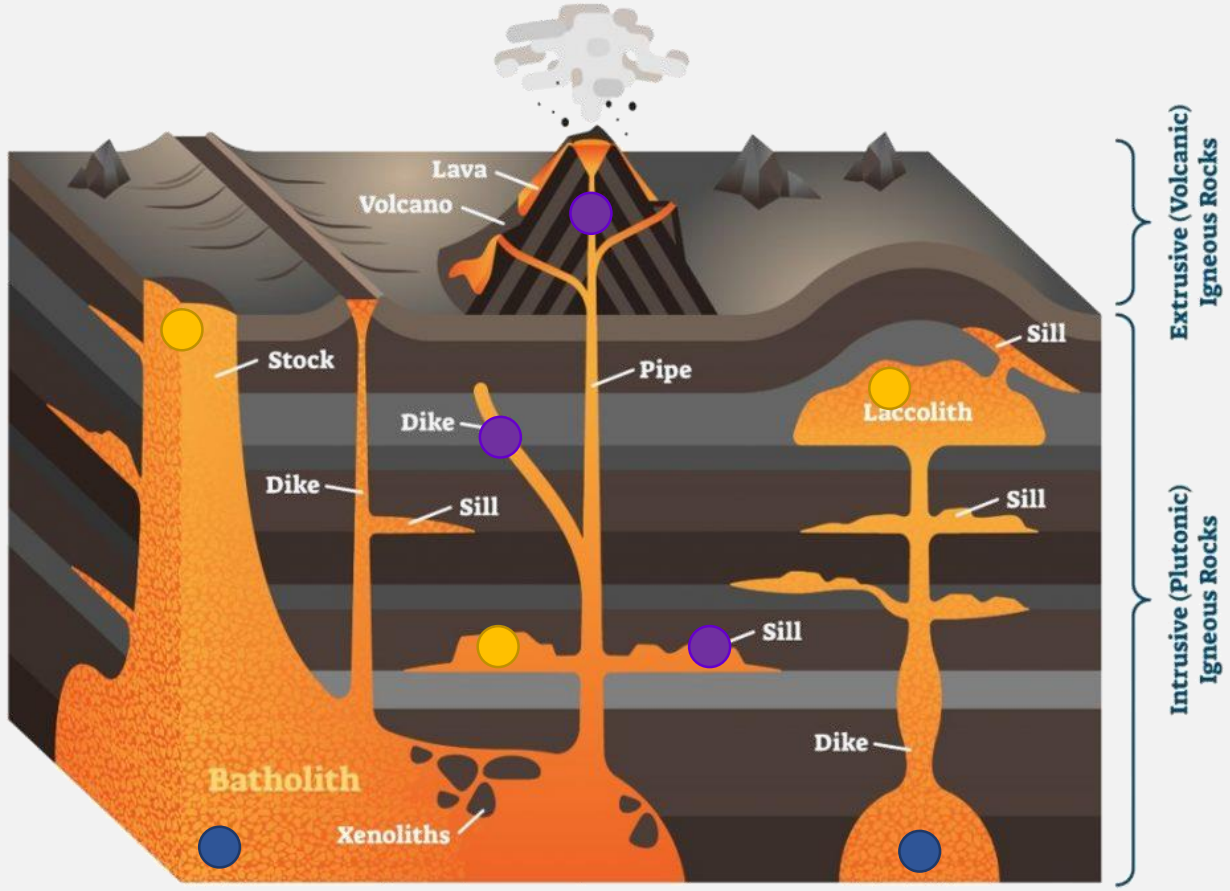
Homogeneous augite crystal



Augite with thin pigeonite lamellae



Orthopyroxene that used to be pigeonite, i.e., “inverted pigeonite”, with thick augite lamellae



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Mafic and silicic intrusions

You should now be able to:

- Describe the crystallization conditions of cumulus and intercumulus minerals
- Recognise cumulate textures in thin section
- Recognise exsolution in pyroxenes, and use the resulting textures to interpret the cooling rate



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Mafic and silicic intrusions

2.1 Mafic intrusions: gabbro and dolerite

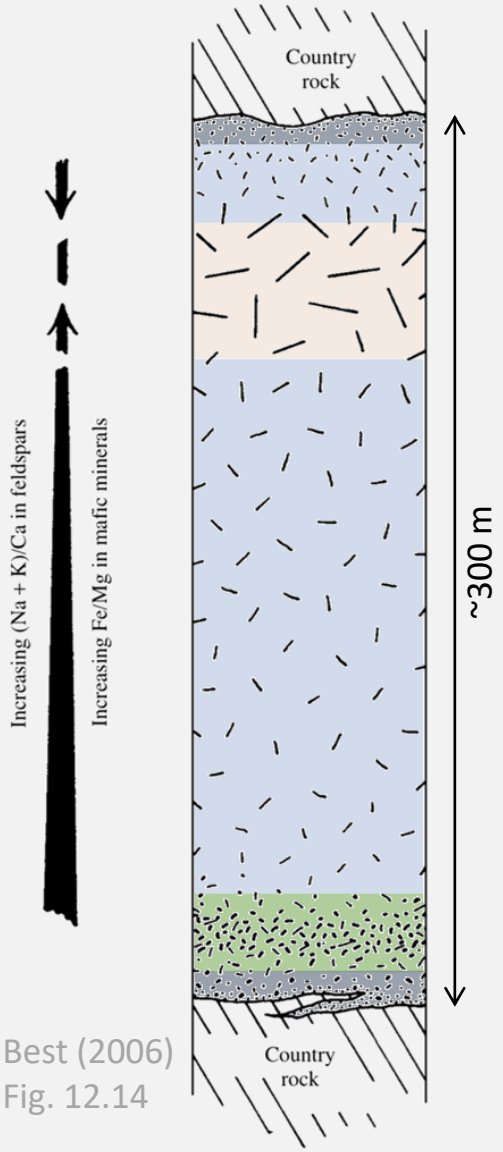
2.2 Closed-system fractional crystallization in a large mafic magma chamber: the Skaergaard layered intrusion

2.3 Crystallization in smaller mafic intrusions: the Palisades Sill

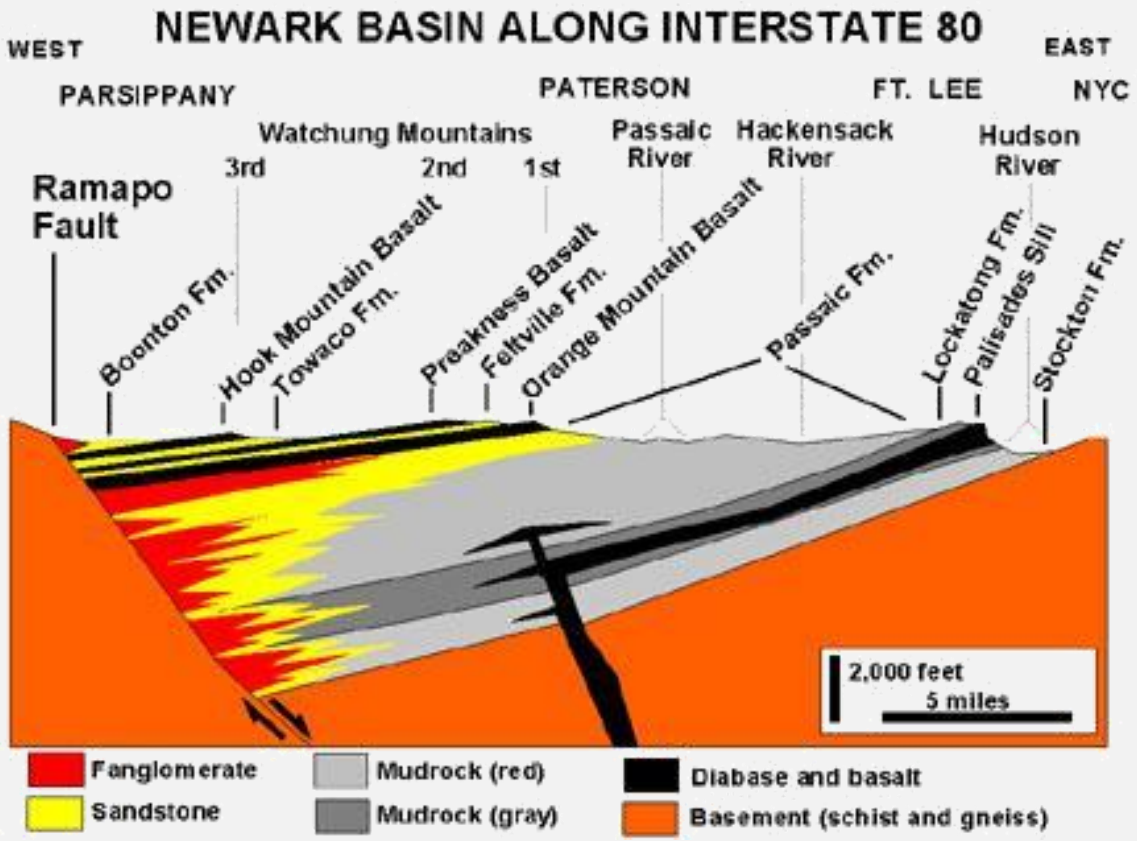


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Crystallization in a smaller mafic intrusion: the Palisades Sill



Best (2006)
Fig. 12.14

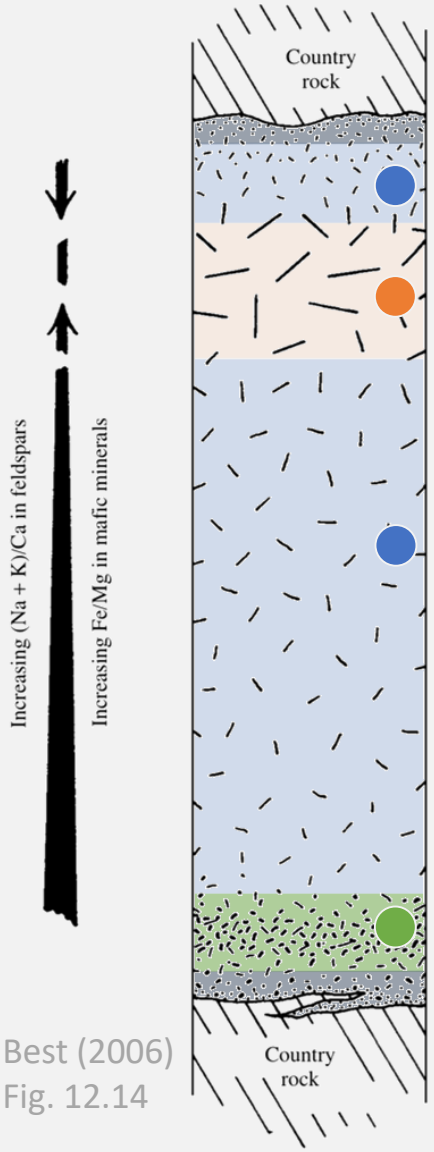




Triassic (~200 Ma) intrusion associated with large-scale rifting during the break-up of Pangaea.

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Palisades Sill: Compositional and modal layering



Best (2006)
Fig. 12.14

← Chilled margin

← Normal dolerite, coarsening downwards

Sandwich horizon

In addition to essential minerals (plagioclase and clinopyroxene):

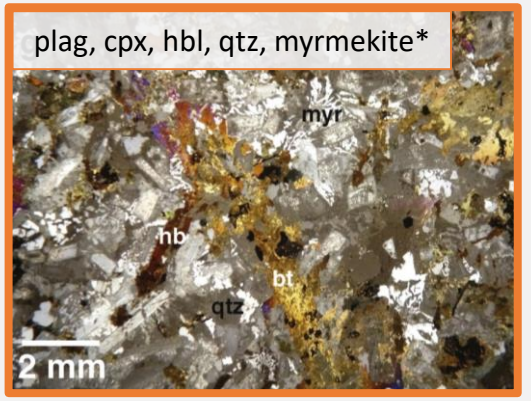
- Up to 30% quartz
- Minor hornblende and biotite
- Myrmekite*

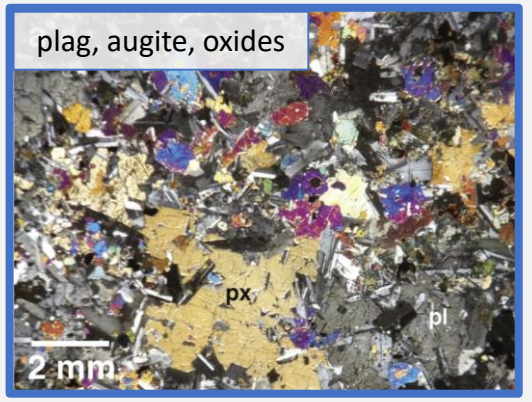
*fine intergrowth of quartz rods inside a plagioclase crystal

← Normal dolerite, coarsening upwards

← Olivine-rich layer: 20-25% olivine Cumulate texture

← Chilled margin Best represents the parental melt composition

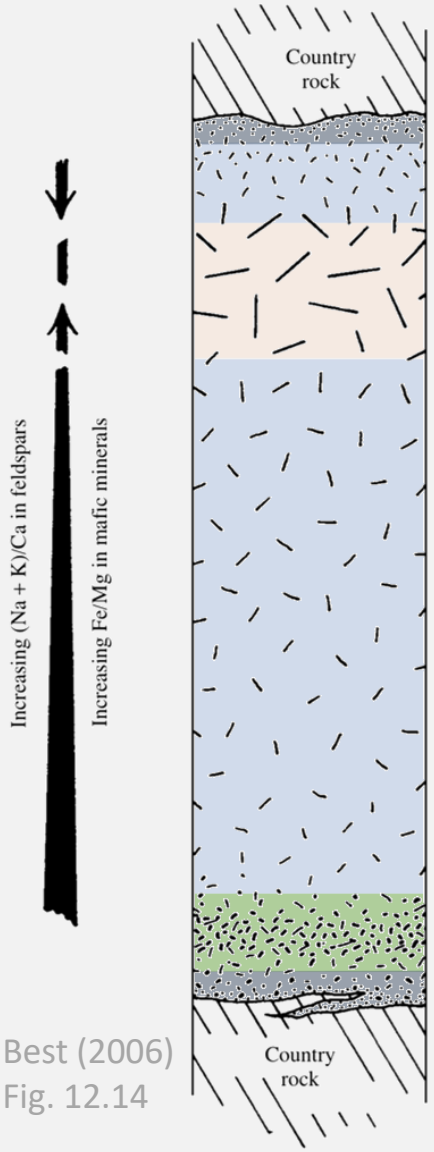






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Palisades Sill: Compositional and modal layering



Best (2006) Fig. 12.14

← Chilled margin

← Normal dolerite, coarsening downwards

Sandwich horizon

In addition to essential minerals (plagioclase and clinopyroxene):

- Up to 30% quartz
- Minor hornblende and biotite
- Myrmekite*

*fine intergrowth of quartz rods inside a plagioclase crystal

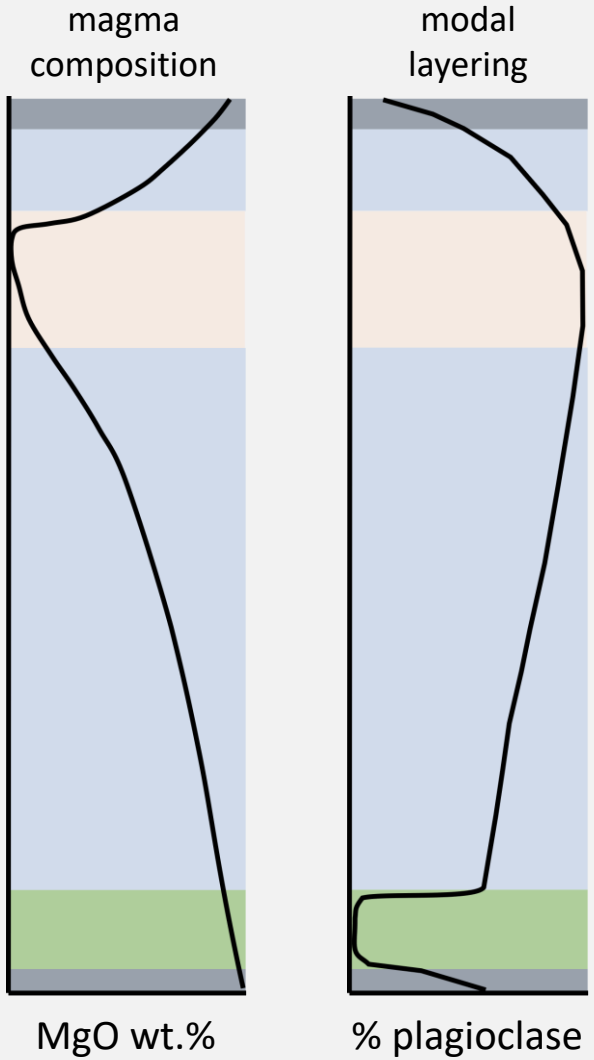
← Normal dolerite, coarsening upwards

← Olivine-rich layer: 20-25% olivine

← Chilled margin

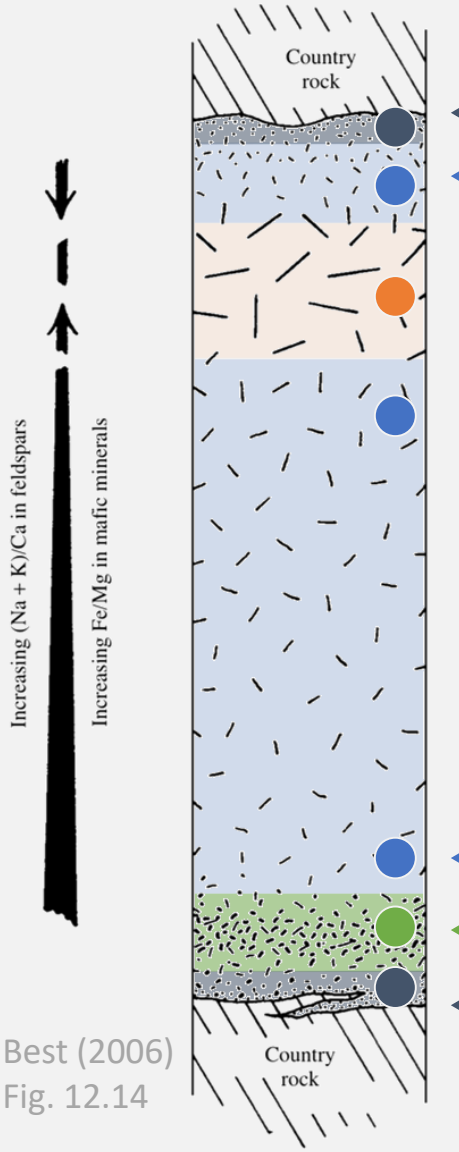
Best represents the parental melt composition

Compositional and modal layering are mostly consistent with closed-system fractional crystallization



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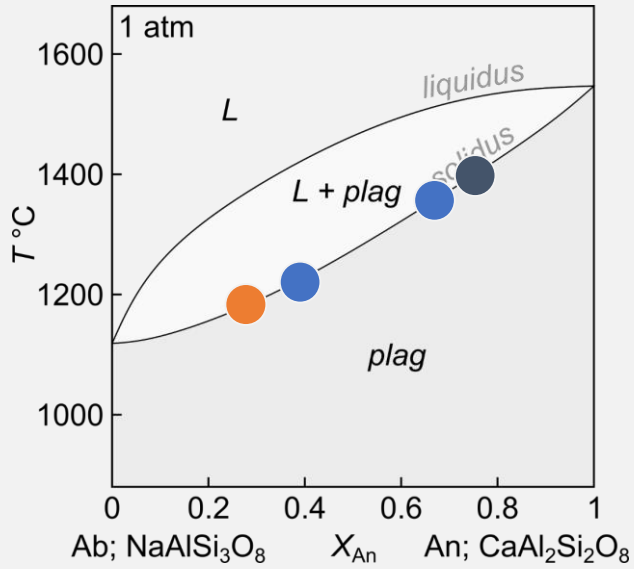
Palisades Sill: Cryptic layering

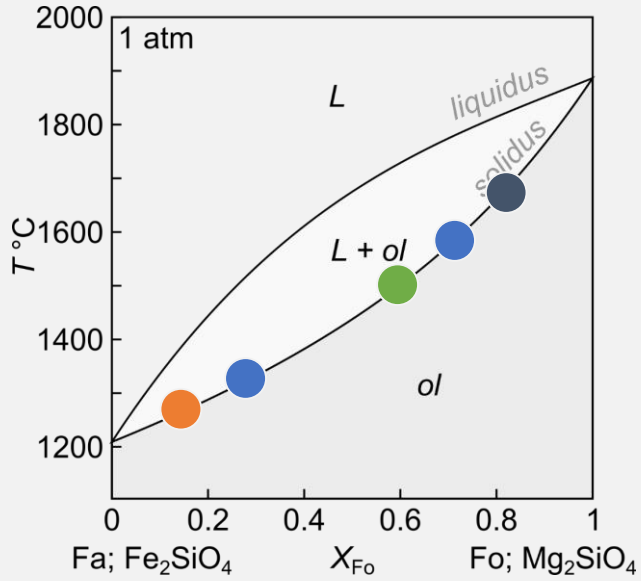


Best (2006) Fig. 12.14

Olivines in the olivine-rich layer are too iron-rich to have crystallized early from the Palisades Sill magma

- These olivines cannot be part of the closed-system fractional crystallization history
- Additional olivine must have been brought into the sill by a second injection of melt





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Mafic and silicic intrusions

You should now be able to:

- Explain why modal and cryptic layering occur in mafic sills, with reference to closed-system fractional crystallization
- Compare and contrast the igneous textures that might be observed in rocks from large layered mafic intrusions versus rocks from smaller mafic sills



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Mafic and silicic intrusions

3.1 Granites, granitoids, and how they are formed

3.2 Igneous textures in granitic rocks

3.3 Granite formation in different tectonic settings

- Continental collision zones
- Continental margins



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Silicic intrusions: Granites and granitoids



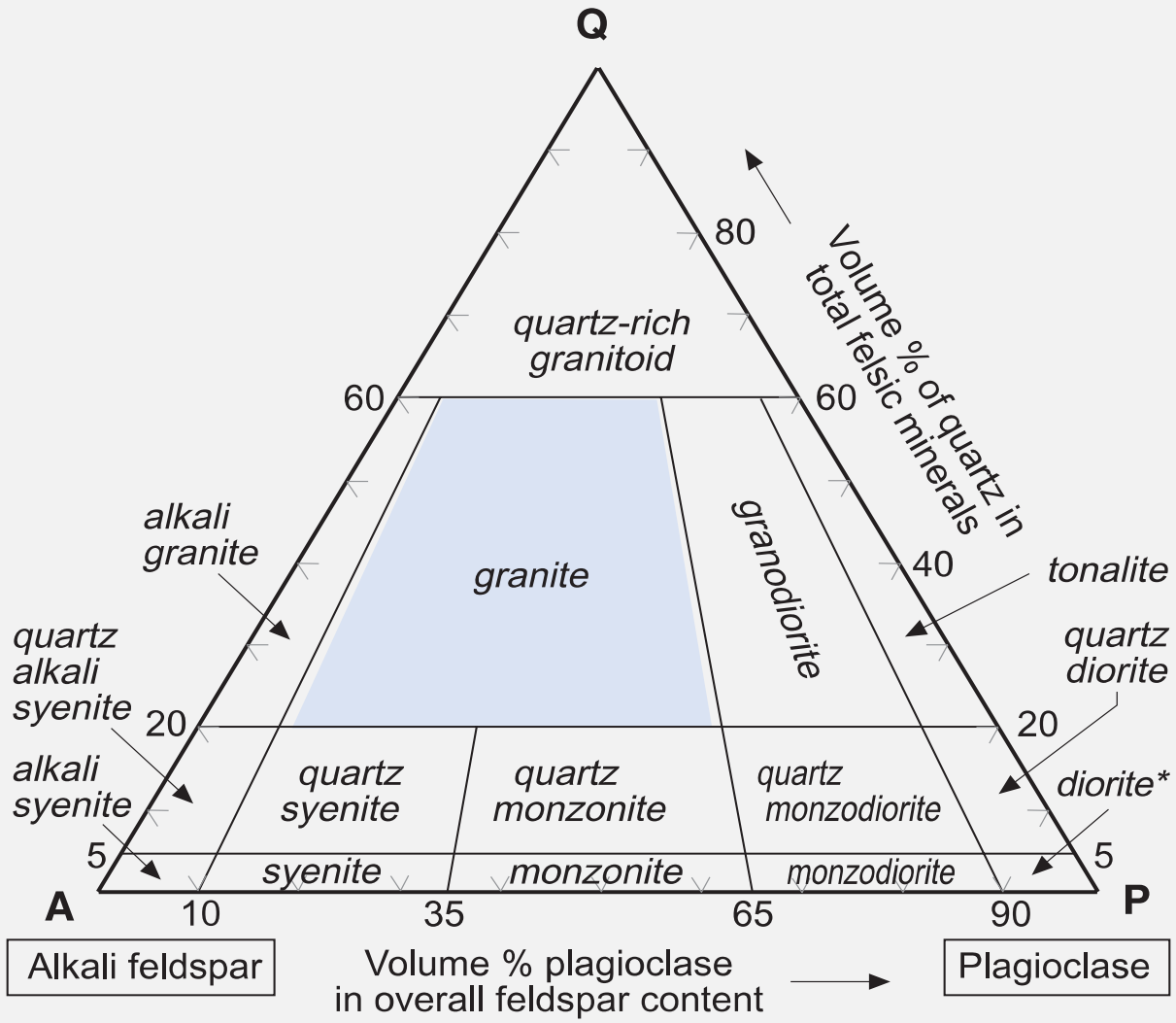
Granites and granitoids:

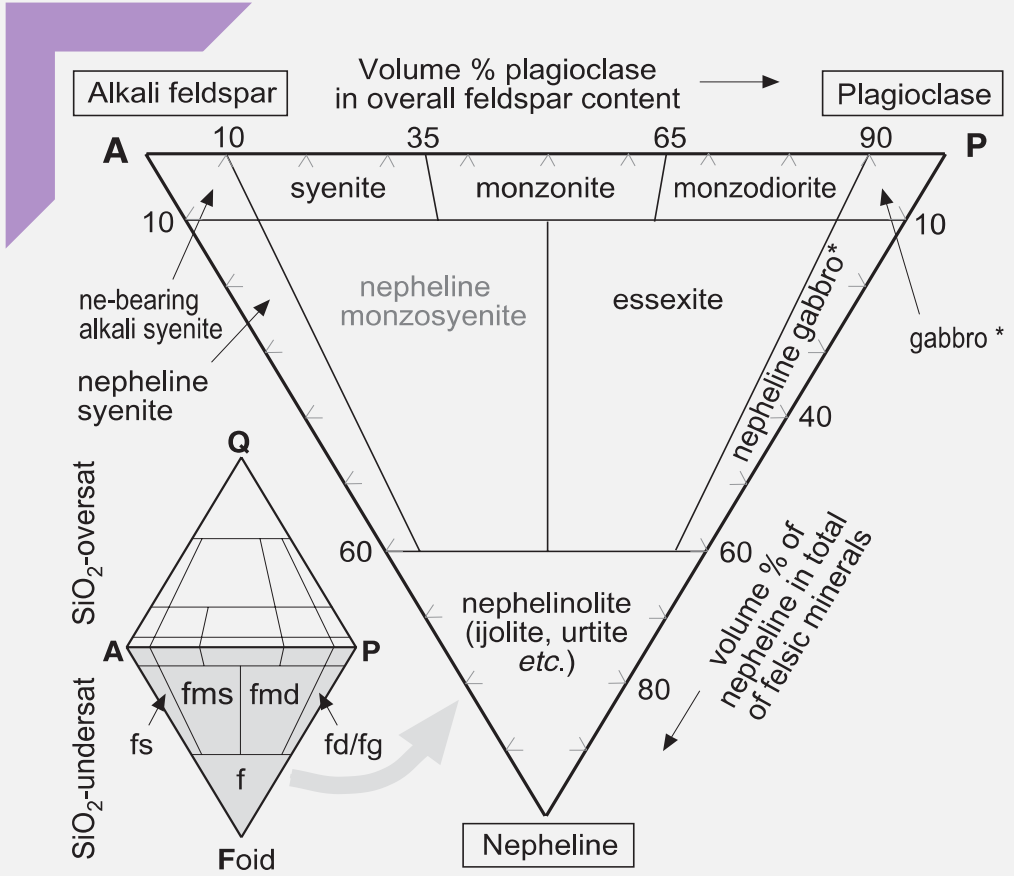
- The coarse- and medium-grained equivalents of andesites, dacites and rhyolites
- The most abundant rock types in the continental crust
- Form mainly at continental arcs and in continental collision zones

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Granites are classified by their modal mineralogy

These diagrams are in the Practical Reference Handout





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Granites are classified by the type of crustal rock they formed from

S-type

Formed by melting of sedimentary crustal rocks



Common in continental collision zones, e.g. Himalaya

I-type

Formed by melting of igneous crustal rocks



Common in continental margins, e.g. Western USA

A-type

Formed in anorogenic settings when the lower crust is melted under dry conditions

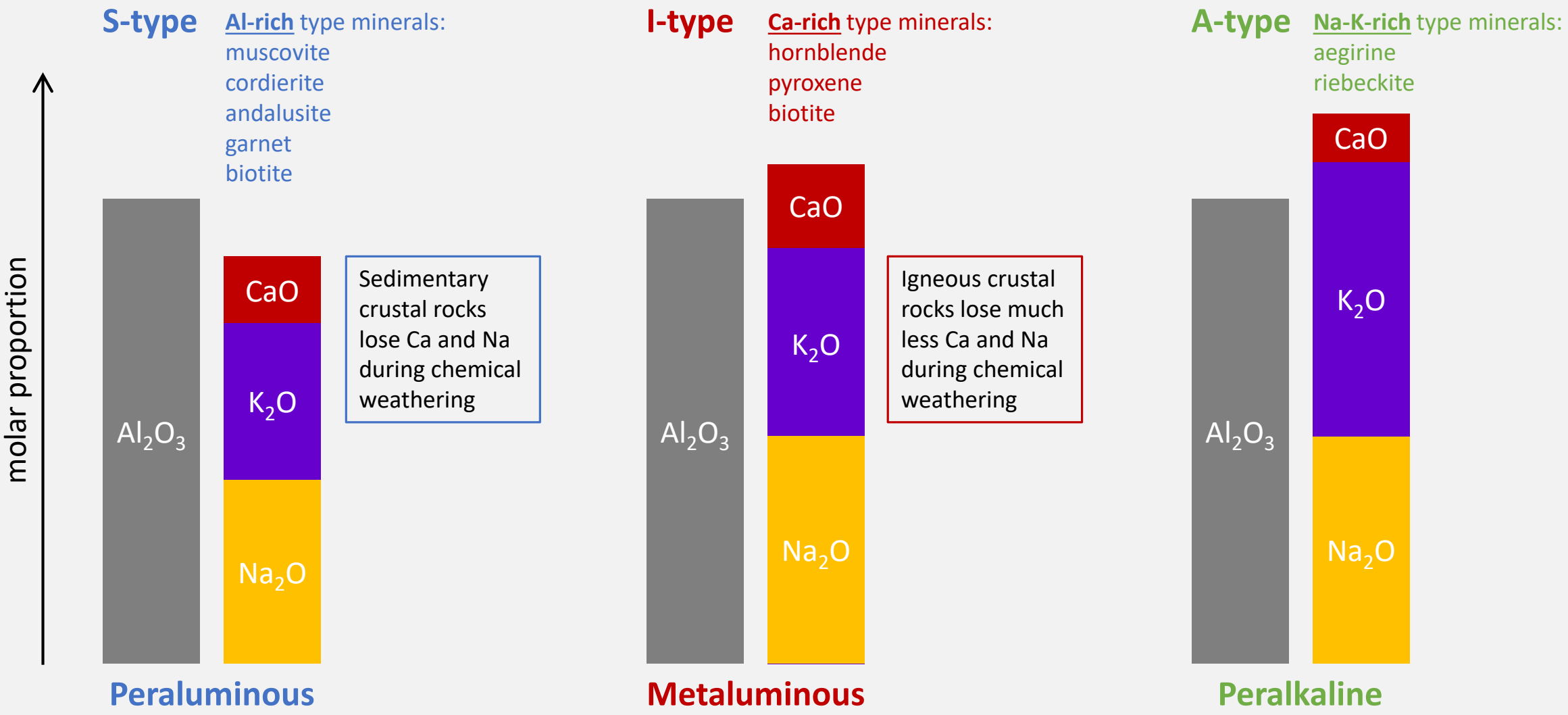


Common in continental rifts, e.g. SE Australia

There are also **M-type** granites which form by extensive fractional crystallization of mantle-derived melt. True M-type granites are relatively rare.

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Granites are classified by their alumina saturation (molar ratio of Al_2O_3 to Na_2O+K_2O+CaO)



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S-type granite: what to look for in a thin section

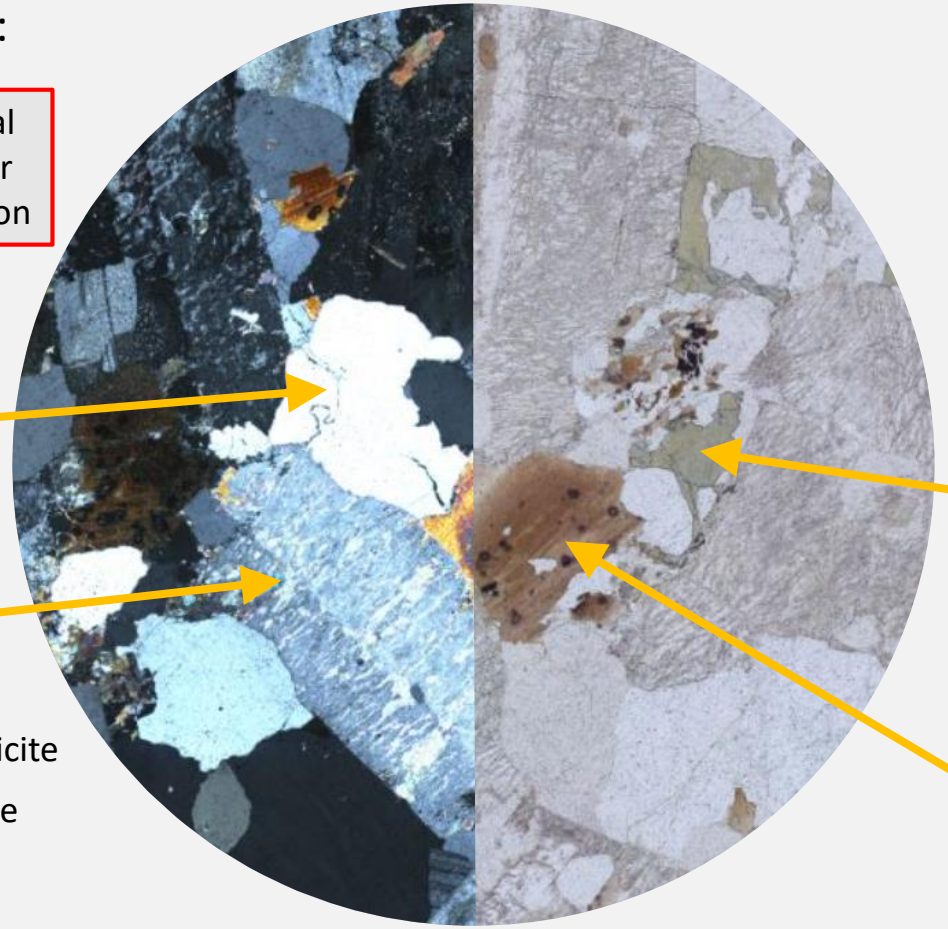
See also the Practical Reference Handout

Essential minerals found in <u>all</u> granites:

- Plagioclase (high-Na)
- Alkali feldspar
- Quartz

Same essential minerals as rhyolite

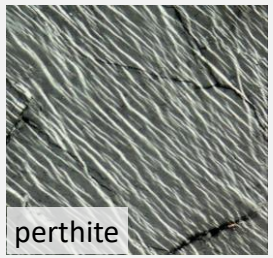
Estimate modal proportions for QAP classification



2 mm

Quartz

fresh, unaltered



Alkali feldspar

simple twinning = <u>orthoclase</u>

often shows dusty alteration to sericite

look for perthite exsolution lamellae

Coarse grain size, granular texture

Type minerals in S-type granites:

- muscovite
- tourmaline
- biotite

And other **<u>Al-rich</u> minerals**, e.g. cordierite

NOT hornblende

Tourmaline

coloured and pleochroic in PPL, moderate relief, no cleavage

Biotite

look for pleochroic haloes in PPL

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S-type granite: what to look for in a thin section

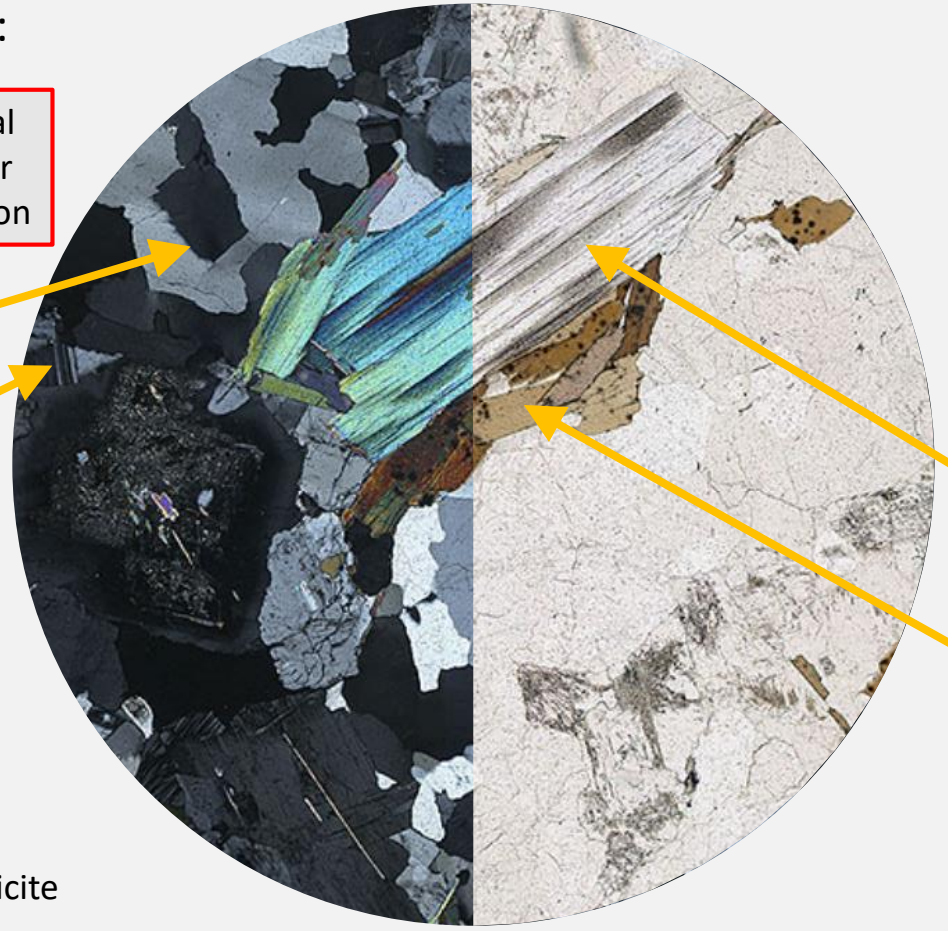
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- Alkali feldspar
- Quartz

Same essential minerals as rhyolite

Estimate modal proportions for QAP classification



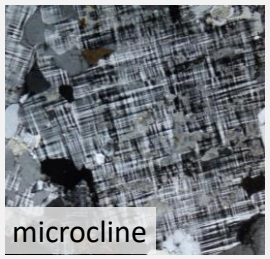
3 mm

Quartz

fresh, unaltered

Plagioclase

look for multiple lamellar twinning



microcline

Alkali feldspar

cross-hatch twinning = microcline

often shows dusty alteration to sericite

look for perthite exsolution lamellae

Type minerals in S-type granites:

- muscovite
- tourmaline
- biotite

And other **Al-rich minerals**, e.g. cordierite

NOT hornblende

Muscovite

Biotite

look for pleochroic haloes in PPL

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I-type granite: what to look for in a thin section

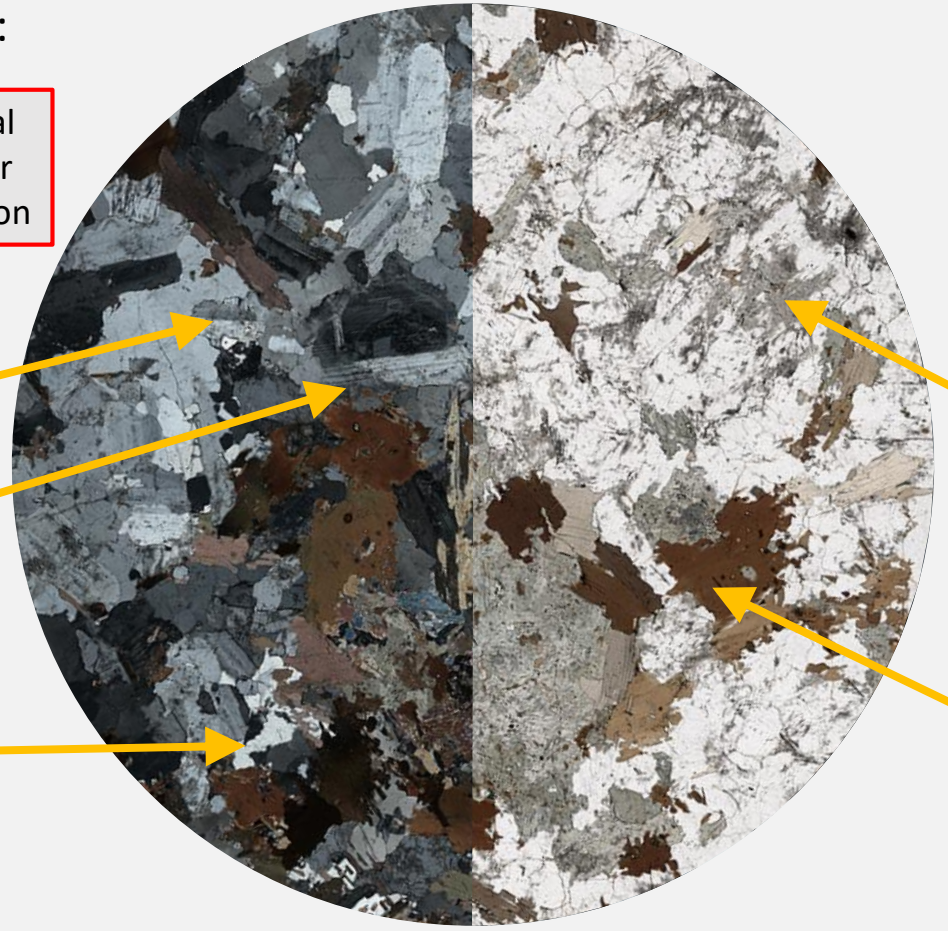
See also the Practical Reference Handout

Essential minerals found in all granites:

- Plagioclase (high-Na)
- Alkali feldspar
- Quartz

Same essential minerals as rhyolite

Estimate modal proportions for QAP classification



Alkali feldspar

simple twinning = orthoclase

Plagioclase

multiple lamellar twinning

Quartz

Coarse grain size, granular texture

Type minerals in I-type granites:

- hornblende
- biotite

And other **Ca-rich minerals**

NOT muscovite or tourmaline

Hornblende

pleochroic in shades of brown and/or green

Biotite

pleochroic haloes less common than in S-type granites

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A-type granite: what to look for in a thin section

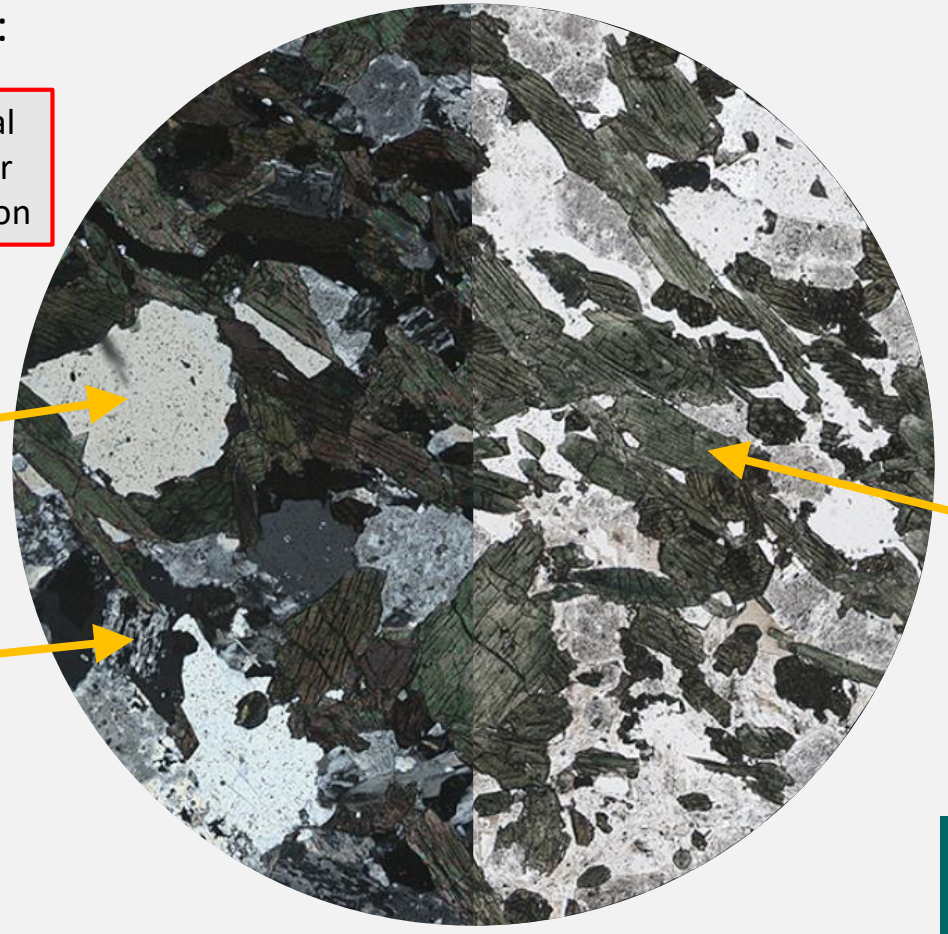
See also the Practical Reference Handout

Essential minerals found in <u>all</u> granites:

- Plagioclase (high-Na)
- Alkali feldspar
- Quartz

Estimate modal proportions for QAP classification

A-type granites often characterized by **abundant alkali feldspar** and **very little quartz**



3 mm

Quartz

fresh, unaltered

Alkali feldspar

<u>orthoclase</u> with perthitic texture and dusty alteration to sericite

Coarse grain size, granular texture

Type minerals in A-type granites:

- alkali pyroxenes e.g. aegirine
- alkali amphiboles e.g. riebeckite
- Fe-rich biotite

And other unusual **<u>Na-K-rich</u> minerals**

Aegirine

distinctive blue-green pleochroic colours in PPL

Sodic pyroxenes and amphiboles are strongly coloured and pleochroic. Typical colours are grass green, blue-green, dark inky blue.

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A-type granite: what to look for in a thin section

See also the Practical Reference Handout

Essential minerals found in <u>all</u> granites:

- Plagioclase (high-Na)
- Alkali feldspar
- Quartz

Estimate modal proportions for QAP classification

A-type granites often characterized by **abundant alkali feldspar** and **very little quartz**

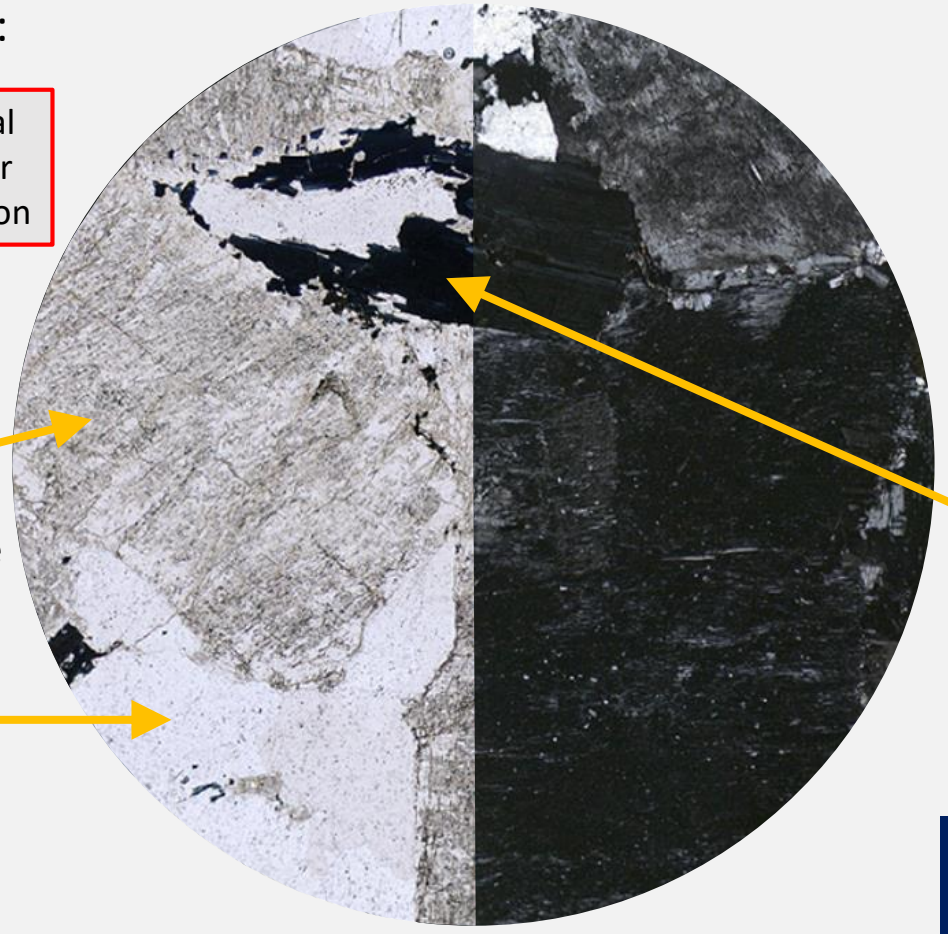
Alkali feldspar

<u>orthoclase</u> with dusty alteration to sericite and thin perthite exsolution lamellae

Quartz

fresh, unaltered

Coarse grain size, granular texture



2 mm

Type minerals in A-type granites:

- alkali pyroxenes e.g. aegirine
- alkali amphiboles e.g. riebeckite
- Fe-rich biotite

And other unusual **<u>Na-K-rich</u> minerals**

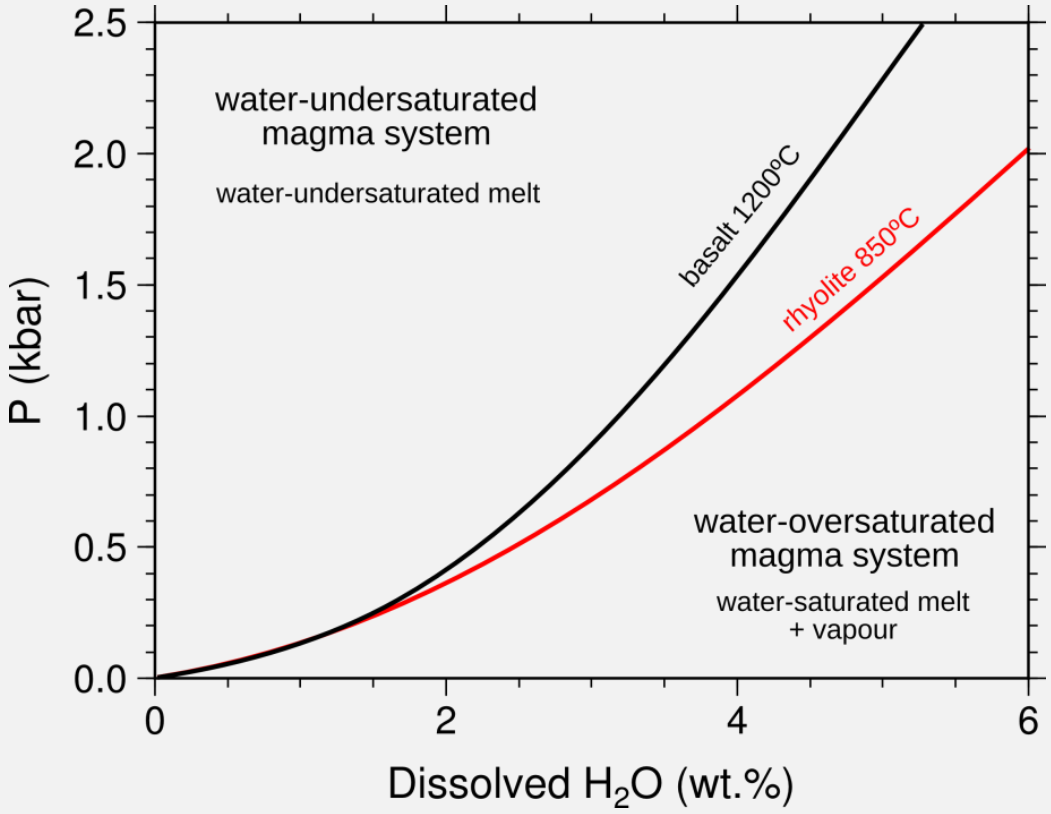
Riebeckite

strongly pleochroic in PPL; this view shows the deep inky blue colour

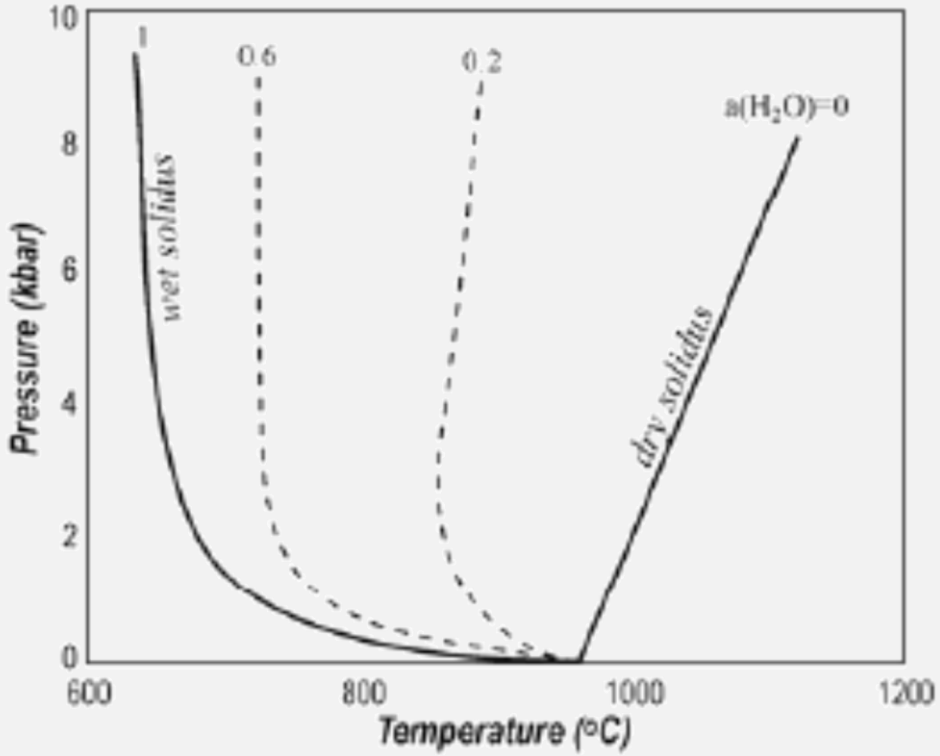
Sodic pyroxenes and amphiboles are strongly coloured and pleochroic. Typical colours are grass green, blue-green, dark inky blue.

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Granitic magmas may have high dissolved water contents



At any given pressure, rhyolitic (granitic) magma can contain more dissolved water than basaltic magma



Rapid decrease in H_2O solubility at 600-700 °C can cause exsolution of large volumes of H_2O

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Mafic and silicic intrusions

You should now be able to:

- Classify granites and granitoid rocks using the QAP ternary diagram
- Explain the origin of S-type, I-type and A-type granites, and outline the main differences in their chemistry and mineralogy
- Classify granites as S-type, I-type or A-type on the basis of observations made on thin sections
- Outline the importance of water in granitic magmas



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Mafic and silicic intrusions

3.1 Granites, granitoids, and how they are formed

3.2 Igneous textures in granitic rocks

3.3 Granite formation in different tectonic settings

- Continental collision zones
- Continental margins



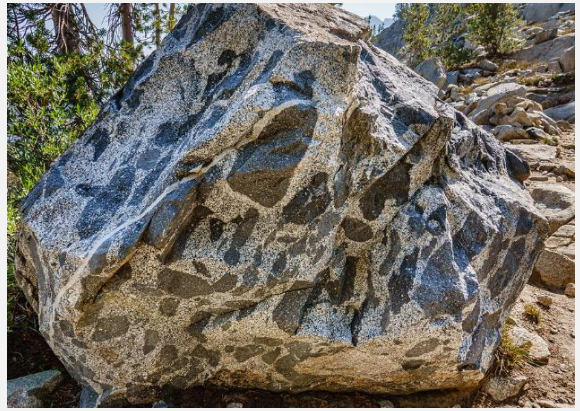
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Outcrop-scale textures: Enclaves, inclusions and xenoliths

crustal xenoliths: sedimentary



crustal xenoliths (igneous)



composite xenoliths





stoping



cognate inclusions



restite xenoliths

Enclaves are cm- to m-sized inclusions that have a contrasting lithology, composition or grain size to their host granite.

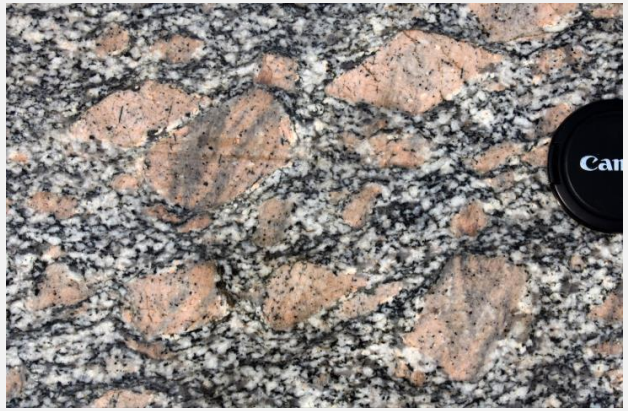
- Xenoliths of country rock – often incorporated by stoping at the roof of the granite intrusion
- Restite: unmelted residue of magma genesis
- Cognate inclusions: igneous material that crystallized from the magma at an earlier stage of fractionation
- Single crystals or crystal cumulates
- Composite xenoliths containing two or more contrasting

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Hand specimen-scale textures: Megacrysts and crystal foliations

megacrysts of white alkali feldspar





megacrysts of pink orthoclase

Alkali feldspar has a low nucleation rate and high growth rate in silicic magmas

Two ways to form alkali feldspar megacrysts:

1) Alkali feldspar is the first mineral to nucleate, and has grown to a considerable size before the other minerals begin to crystallize
2) All minerals nucleate at about the same time, but alkali feldspar grows fastest and forms the largest crystals

foliation





foliation

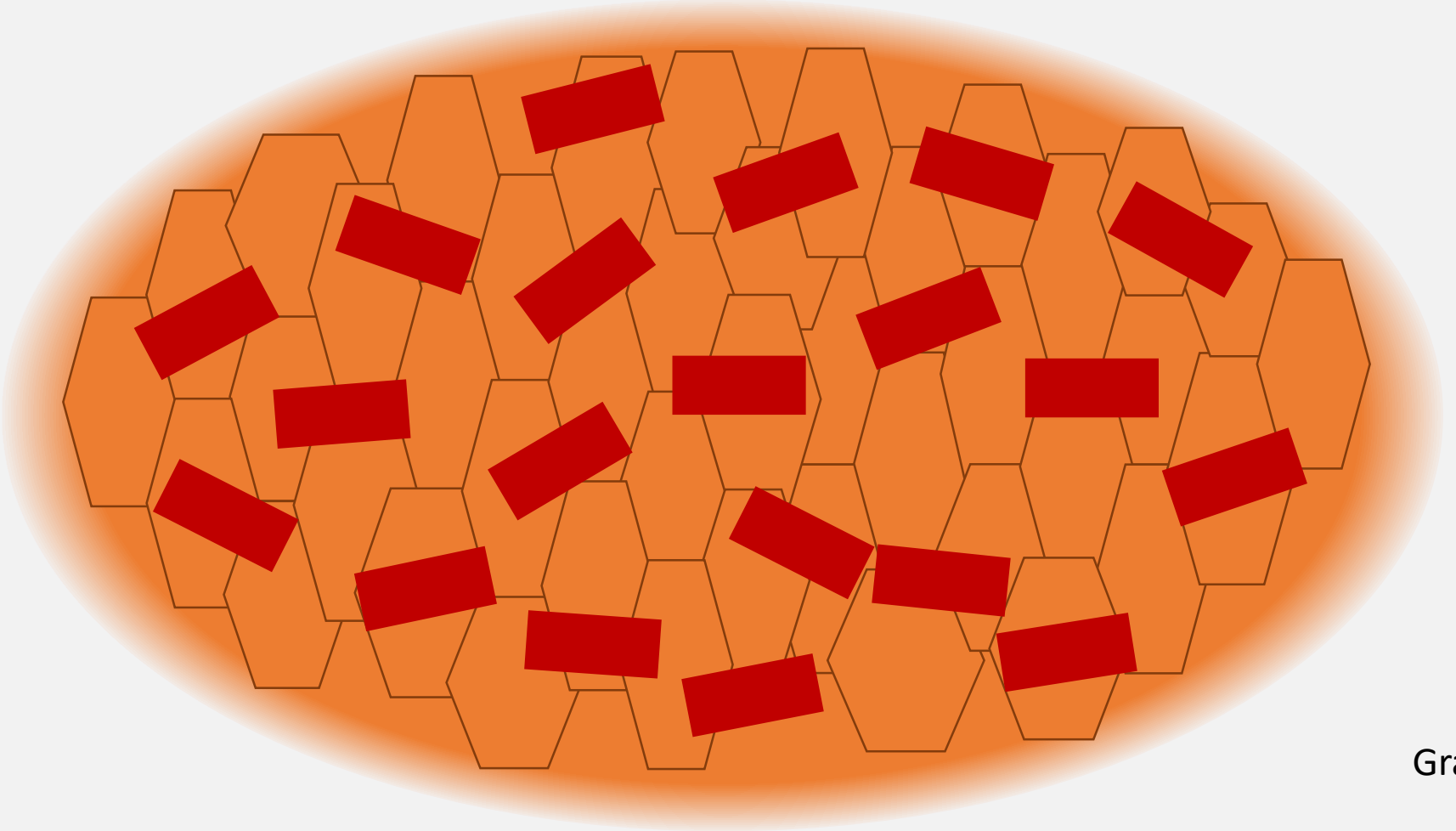
Granites are too viscous to form cumulate textures or rhythmic layering

Mineral alignments and foliations in granites can be created by:

- Syn-crystallization shearing: intrusion of new magma into a pre-existing granitic magma can push already-formed crystals into a preferred alignment
- Post-crystallization tectonic deformation: foliations typically form perpendicular to the maximum principal stress direction

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“Magmatic sedimentation” does not occur in silicic magmas



1. Large, euhedral crystals grow free-floating in the magma

2. High-silica magma is too viscous for crystal settling to occur

No crystal accumulation at the floor of the chamber

3. The rest of the magma solidifies

Early-formed euhedral crystals randomly distributed in the rock

Granites do not form cumulate textures

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Mafic and silicic intrusions

You should now be able to:

- Describe the range of textures that can be observed in granitic rocks, from outcrop scale to thin section scale
- Explain why cumulate textures do not form in granitic magmas



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Mafic and silicic intrusions

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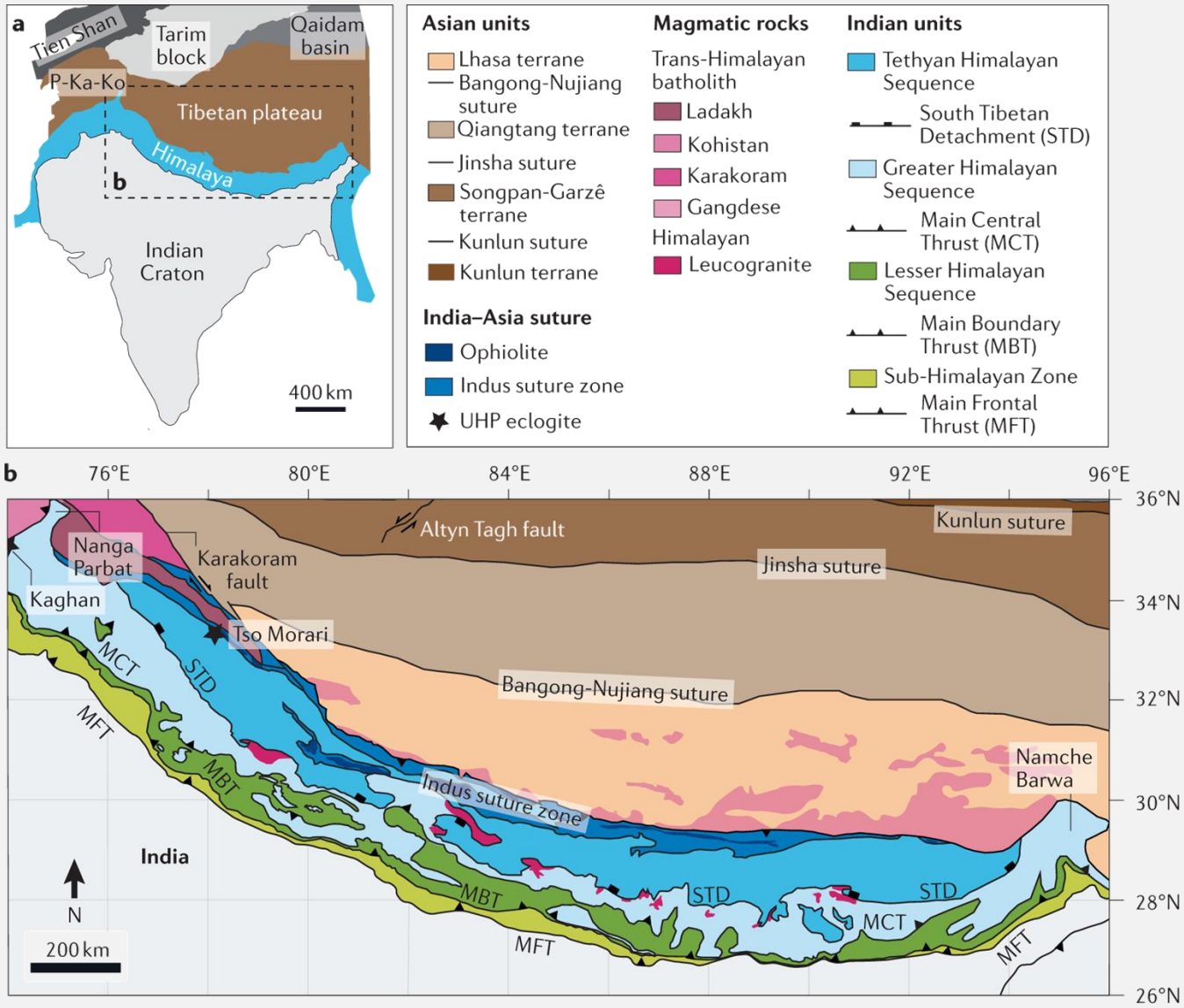
- Continental collision zones

- Continental margins



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Granites at continental collision zones: the High Himalayan leucogranites



The Trans-Himalayan batholith was intruded in a continental arc setting during subduction of the Tethys Ocean beneath Asia

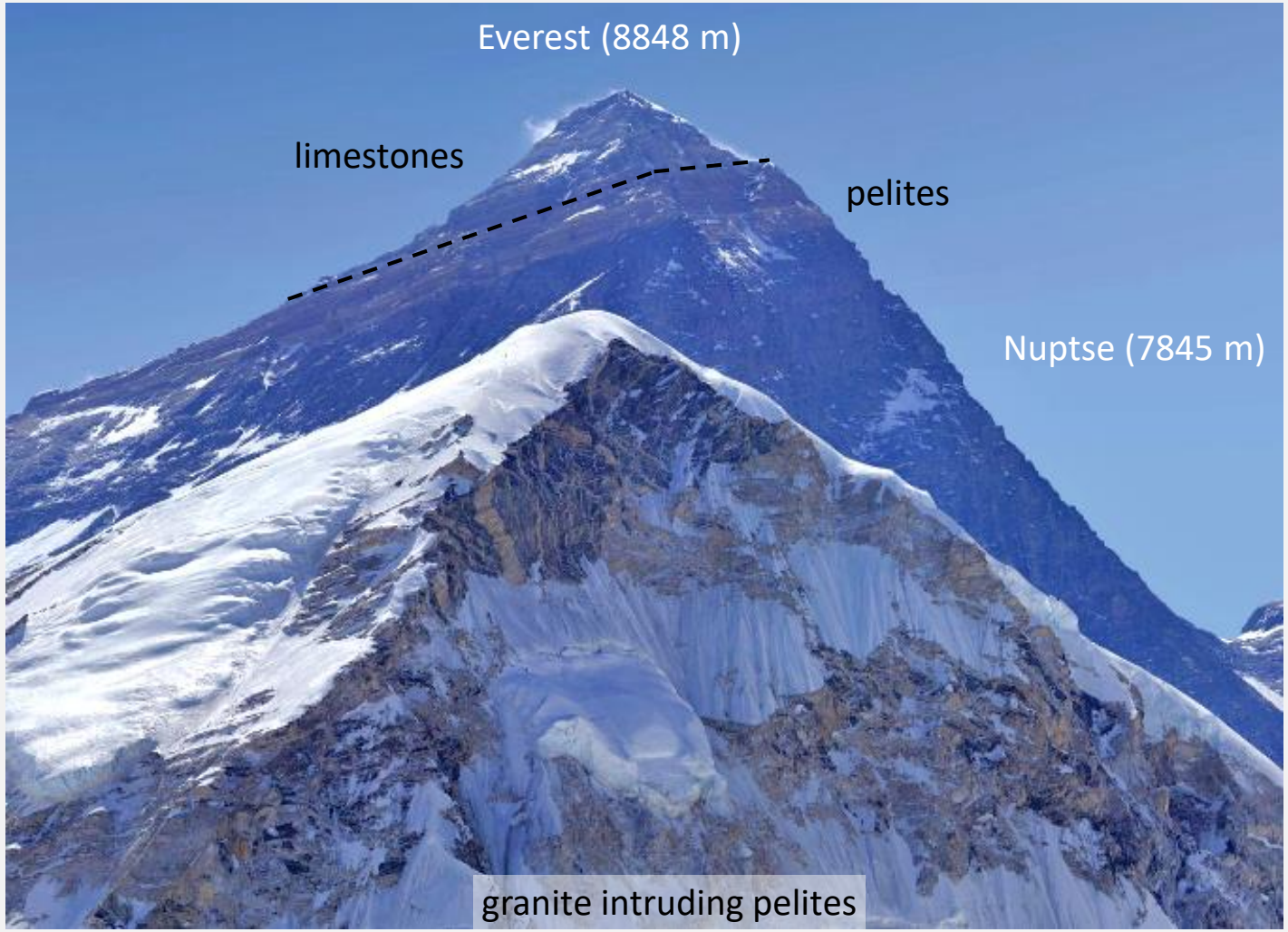
- Intruded ~130-55 Ma
- Mainly I-type granites with calc-alkaline signatures

The High Himalayan leucogranites were intruded in a continental collision setting, into rocks undergoing metamorphism

- Intruded ~25-15 Ma
- Exclusively S-type granites
- $^{87}Sr/^{86}Sr$ ratio is consistent with a sedimentary

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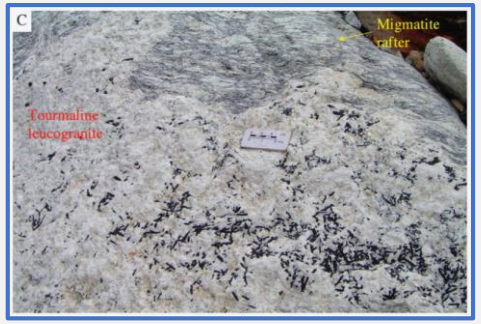
Granites at continental collision zones: the High Himalayan leucogranites



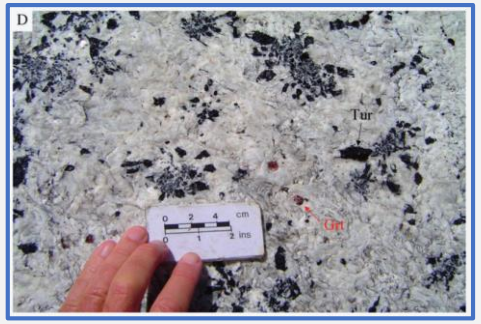
granite intruding pelites



cordierite



tourmaline



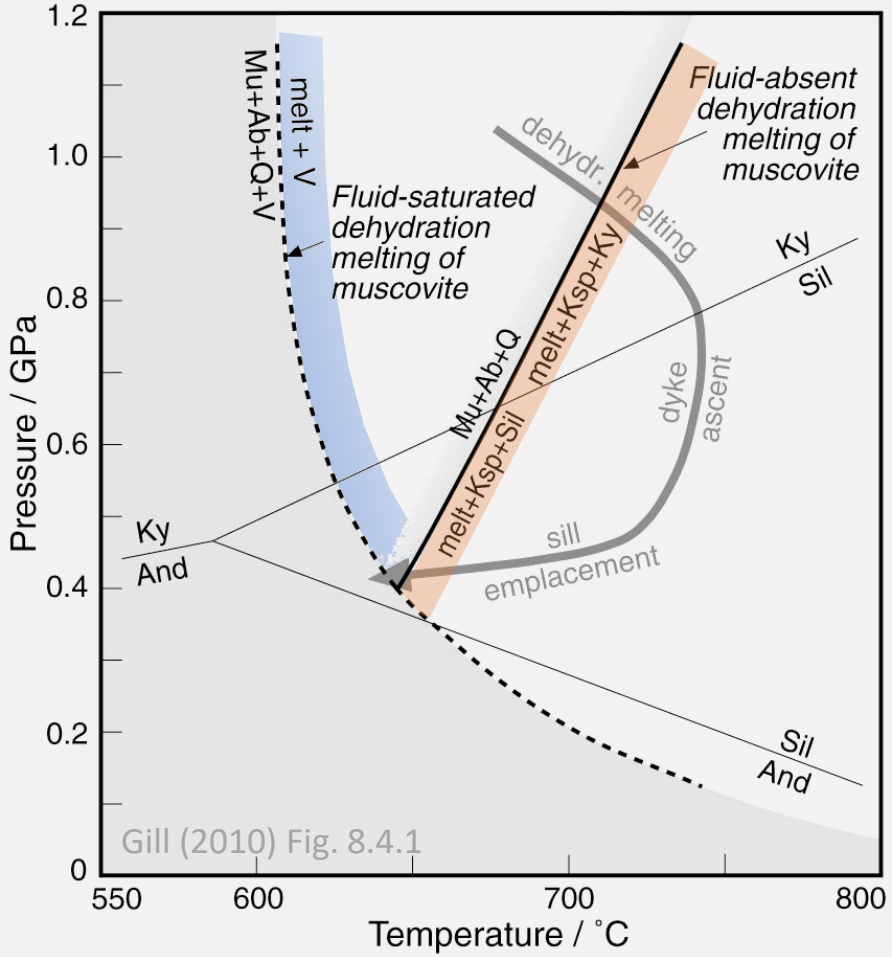
tourmaline + garnet

Peraluminous (S-type) compositions formed by melting Al-rich pelites (metamorphosed mudstones).

Al-rich type minerals.

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Formation of S-type granites in continental collision zones



Pelites and metapelites partially melt via reactions involving muscovite

Fluid (vapour) present:

$\mathbf{mu} + qz + plag + H_2O \rightarrow melt$ melt fraction F ≈ 0.4

Fluid (vapour) absent:

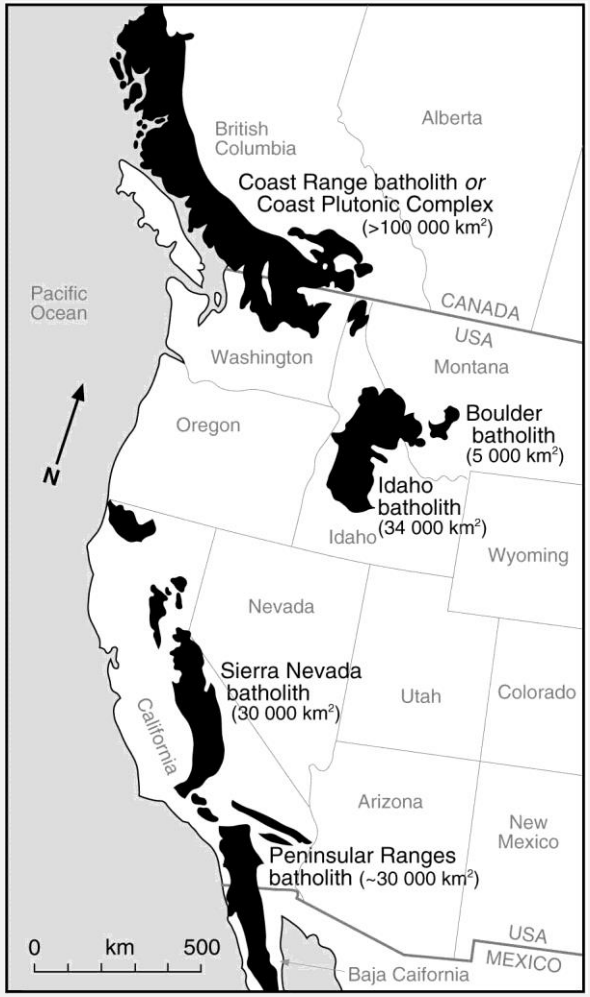
$\mathbf{mu} + qz + plag \rightarrow melt$ melt fraction F ≈ 0.12

Crustal melting can occur during heating and/or decompression

The resulting granitic magmas have high $^{87}Sr/^{86}Sr$ ratios consistent with melting of continental crust

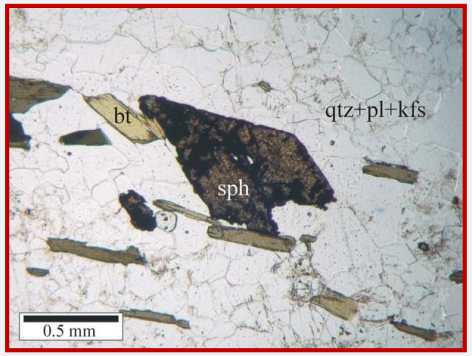
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Granites at continental margins: the Cordilleran granites, western USA





Titanite (sphene) + biotite



Biotite with pleochroic haloes



The Cordilleran granites intruded in a continental arc setting during subduction beneath the North American Plate:

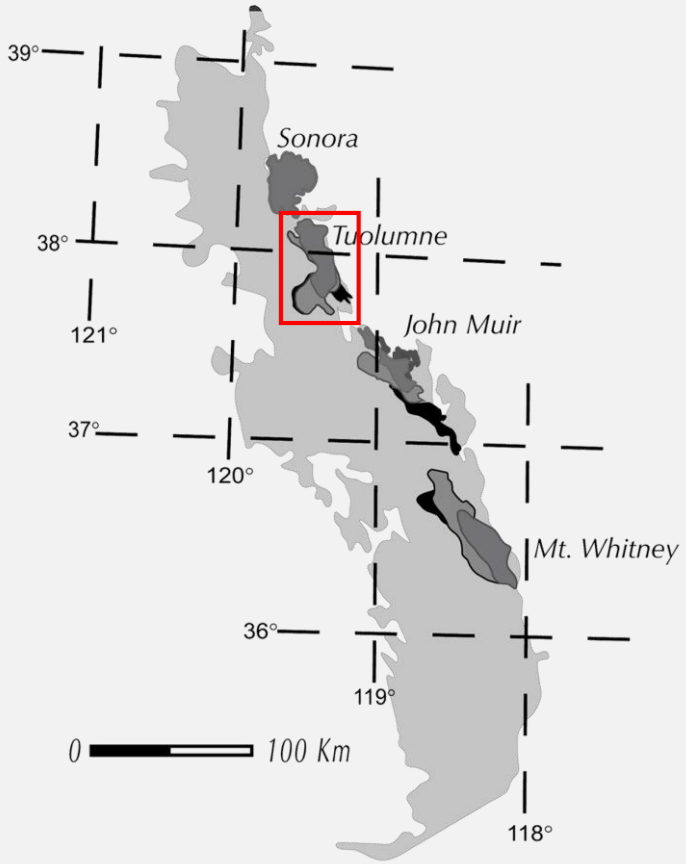
- Sierra Nevada batholith intruded ~114-87 Ma
- Idaho batholith in intruded ~98-53 Ma
- I-type tonalites, granodiorites and granites

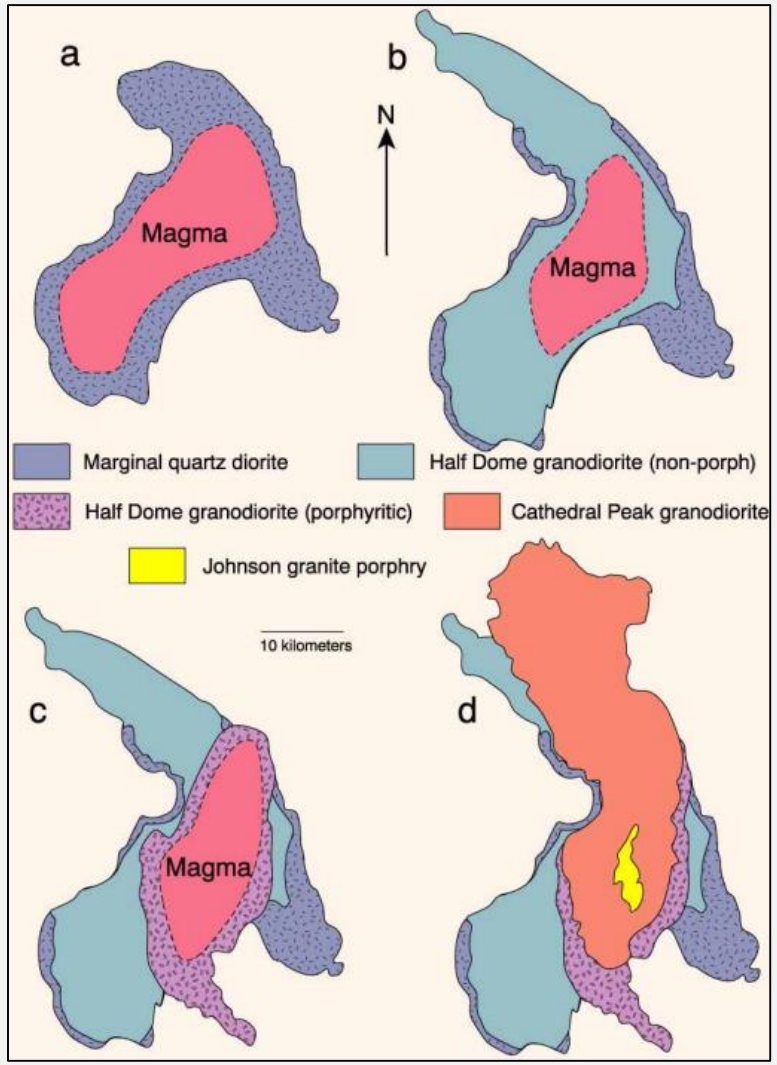
Metaluminous (I-type) compositions formed by melting Al-poor, Ca-rich igneous rocks in the lower crust.

Al-poor, Ca-rich type minerals.

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The Sierra Nevada batholith: a series of composite, zoned plutons





Tuolumne Intrusive Series, Yosemite National Park:

(a) Initial intrusion: quartz diorite

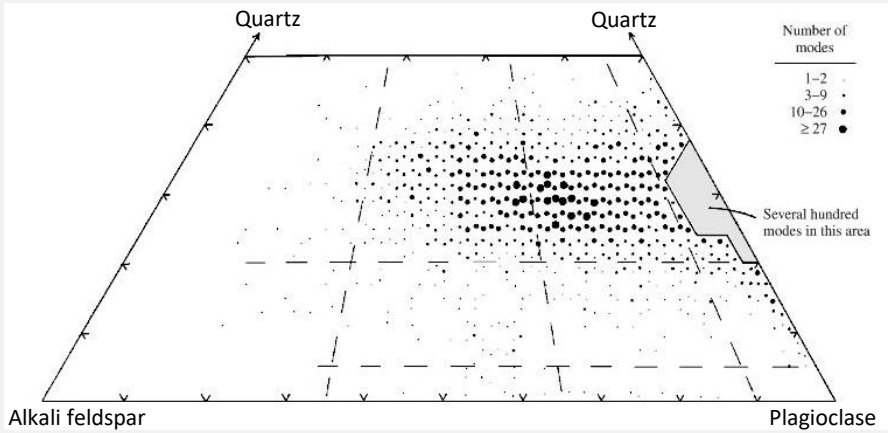
(b) Fresh magma intrusion 1: granodiorite

(c) Fresh magma intrusion 2: porphyritic granodiorite

(d-1) Fresh magma intrusion 3: granodiorite

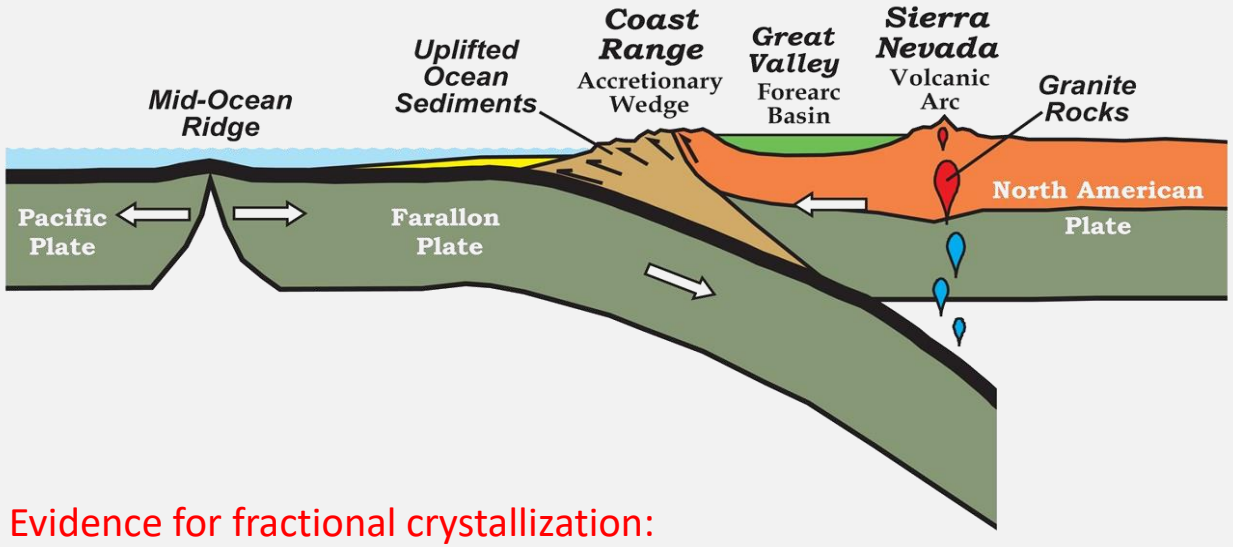
(d-2) Final stage: Granite porphyry

Pluton interiors are typically younger and more felsic than the exteriors



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Formation of I-type granitic magmas at continental margins



Evidence for fractional crystallization:

Zoned plutons represent stages of fractionation. Early intrusions are tonalites; later intrusions are granites.

Evidence for melting of continental crust:

$^{87}Sr/^{86}Sr$ increases with distance from the magmatic arc - indicates greater assimilation of continental crust

Stages of forming an I-type granitic magma:

1. Partial melting of wet peridotite in the mantle wedge produces mafic magmas
2. Mafic magmas crystallize in the lower crust: gabbros and diorites
3. Later intrusions of hot mafic magma cause partial melting of the igneous lower crustal rocks, producing intermediate and evolved magmas
4. Intermediate/evolved magmas ascend to upper crust and form batholiths.
    - Fractional crystallization creates tonalites to granites
    - Variable assimilation of continental crust modifies the isotopic signature

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The Sierra Nevada batholith and the California Gold Rush (1848-49)



Sierra Nevada batholith hosts gold lode deposits:

- Gold is found within quartz veins, faults, and shear zones
- Hot, mineral-saturated fluids mobilized gold from elsewhere in the crust and concentrated it into economically viable deposits
- Granitic magmas supply the hot hydrothermal fluids

The origin of the Sierra Nevada gold is thought to be ancient oceanic terranes now accreted onto the North American Plate. The original mineralization probably occurred at submarine volcanic vents akin to black smokers.



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Mafic and silicic intrusions

You should now be able to:

- Compare and contrast the granites produced in different tectonic settings, with reference to specific examples
- Use mineralogical evidence from hand specimens and thin sections to interpret the tectonic setting of formation of granites
- Outline the importance of water in creating economically viable ore deposits around granite intrusions



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